

Chapter 20. Extracorporeal Life Support for Cardiopulmonary Failure

Heidi J. Dalton; Pamela C. Garcia-Filion

Definition and History

Extracorporeal life support (ECLS) is defined as total or partial diversion of a patient's circulating blood volume into a device that can act as a "lung," providing oxygenation and carbon dioxide removal, and a pump, which can act like a "heart," providing circulatory support, or a combination of both functions for patients suffering from both cardiac and pulmonary failure. Adapted from heart-lung machines used for cardiopulmonary bypass, this temporary form of support was termed extracorporeal membrane oxygenation (ECMO) in the past.¹⁻⁴ Because of vast changes in the circulatory devices now available and the patient populations being treated, however, the term ECMO is being replaced by ECLS as the designation for extracorporeal life support of a variety of indications and devices. Both are used interchangeably in the chapter. [Figure 20-1](#) shows an example of an early cardiopulmonary bypass circuit.

Figure 20-1



Source: Tobin MJ: Principles and Practice of Mechanical Ventilation, 3rd Edition: www.accessanesthesiology.com
Copyright © The McGraw-Hill Companies, Inc. All rights reserved.

Early cardiopulmonary bypass circuit. (Courtesy of Heidi Dalton MD, with permission.)

ECLS was first developed to provide respiratory support in premature infants who had inadequate lung development to support gas exchange compatible with life. Difficulties with intracranial hemorrhage secondary to the heparinization required to prevent clotting of the artificial membrane lung and the circuit, however, caused ECLS to quickly fall out of favor in this population. Work continued, and following the development of the silicone-membrane oxygenator by Kolobow and others, ECLS was successfully applied to term newborns with respiratory failure.^{5,6} One of the first of these patients was a newborn named Esperanza (which means hope) by the neonatal intensive care unit nursing staff who were caring for her at the University of California at Irvine in 1976. Esperanza had hypoxemia secondary to what we now would call persistent pulmonary circulation of the newborn. Dr. Robert Bartlett and colleagues were doing research on ECMO at the University of California at Irvine and were contacted for assistance with Esperanza. Their laboratory device was quickly approved by the institutional review board for emergency use, and adapted and applied in an attempt to save Esperanza. The procedure worked well and Esperanza is now a grown woman with children of her own. Following this event, other centers reported success in the use of ECLS in newborns with respiratory failure from diseases such as meconium aspiration syndrome, persistent pulmonary hypertension, and congenital diaphragmatic hernia.⁷⁻¹⁰

This initial success was met with some skepticism, and a randomized controlled trial of ECMO or ECLS versus conventional ventilation was undertaken in newborns with persistent pulmonary hypertension in the late 1980s. This was a two-phase project: if excessive mortality or benefit was noted in one [Loading \[Contrib\]/a11y/accessibility-menu.js](#) be treated with the more successful therapy.¹¹ The safety monitoring board stopped Phase 1 of the study when

four of the ten patients in the conventional arm died while all nine of the patients in the ECMO arm survived. Of the next twenty patients (all receiving ECMO), nineteen patients survived. Thus, 97% (twenty-eight of twenty-nine) of the ECMO patients and 60% (six of ten) of the conventional ventilator patients survived ($p < 0.05$). This trial aroused much controversy, especially regarding the randomization scheme and the fact that only families of patients randomized to ECMO were asked for consent.¹² Another randomized study from the United Kingdom also favored ECLS in neonatal respiratory failure.¹³ Today, ECLS is an accepted therapy for neonates with refractory respiratory failure.¹⁴

At the same time, adults with respiratory failure were also receiving ECLS.^{15,16} One of the first case reports, by Hill in 1972, involved a young man who suffered injuries, including a ruptured aorta, as the result of a motorcycle accident. Supported by a cumbersome ECMO circuit for 3 days, the man survived.¹⁷ Further case reports and a historic meeting regarding ECLS held in Copenhagen in the late 1970s led to a National Institutes of Health-sponsored trial of ECMO versus conventional therapy in adult respiratory failure. Published in 1979 in the *Journal of the American Medical Association (JAMA)*, the ECMO and conventional therapy groups had equivalently poor survival ($<10\%$), and major complications of bleeding.¹⁸ The study also took place during an epidemic of influenza, which may have skewed results. The study, in retrospect, was flawed given lack of consensus definitions of conventional treatment, maintenance of high levels of mechanical ventilator support in both groups with no attempt at lung-protective ventilation in the ECMO patients, and imprecise definitions of organ failure. Of a planned 300-patient enrollment, the study was stopped at ninety-two patients because of the high mortality rates in both groups.

These dismal results dampened enthusiasm for ECLS in the adult patient and it became poorly accepted among adult clinicians in the United States. Across the globe, however, investigators in Italy, Sweden, Germany, Japan, and elsewhere continued work on supporting adult respiratory failure with ECLS techniques.¹⁹ Research by Gattinoni added to the understanding of the pathophysiology of adult respiratory failure, the detrimental impact of mechanical ventilation at high levels on the lung and the potential protective role of ECLS in lung recovery.²⁰ When Gattinoni published results of a low positive pressure ventilation scheme combined with ECLS, primarily for carbon dioxide removal, which achieved 49% survival in patients with similar entry criteria as the failed National Institutes of Health trial, interest in adult ECLS again began to arise. Further studies with similar results from other centers began to appear and culminated with a conference on ECLS in Marburg, Germany, in 1988.^{21,22}

As a means of sharing information regarding patients receiving ECLS, an informal gathering of clinicians, bench researchers, industry representatives, technicians, and survivors was held in 1989. Titled the Extracorporeal Life Support Organization (ELSO), this volunteer consortium has established an international database that contains information on more than 45,000 patients from more than 100 centers. The annual symposium continues to provide dissemination of technology, networking, and discussions of advancements and areas of needed work in the field, and provides an important forum for interaction between clinicians, technicians, and industry and bench researchers.^{23,24} ELSO has also established practice guidelines and standards of care for patients receiving ECLS. Membership in ELSO gives access to database information for research or center inquiry for specific patient populations. Quarterly reports on both individual and collective center outcomes, complications, equipment, and other parameters are provided. ELSO also provides small levels of grant support to advance the field of ECLS. Centers providing ECLS who do not report results to the ELSO group weaken the ability to track ECLS changes and outcomes globally. Currently undergoing some revision, to refine severity of illness parameters and other specifics, the ELSO registry continues to provide the most comprehensive information regarding trends in ECLS on an ongoing basis. Further information regarding this group can be found at www.ELSO.med.umich.edu.

As with other groups, use of ECLS in pediatric respiratory failure patients has been controversial. Attempts to establish mortality risks and outcome in pediatric respiratory failure were a prelude to a randomized trial of ECLS versus conventional care.²⁵ The earlier report of Timmons et al on 470 patients from forty-one pediatric intensive care units with respiratory failure (defined as a fractional inspired oxygen concentration [FiO_2] ≥ 0.50 and positive end-expiratory pressure [PEEP] ≥ 6 cm H₂O for ≥ 12 hours) noted a mortality rate of 43%.²⁶ Using variables that included age, operative status, Pediatric Risk of Mortality (PRISM) score, FiO_2 , respiratory rate, peak inspiratory pressure, PEEP, arterial oxygen tension (PaO_2), and arterial carbon dioxide tension ($PaCO_2$), a pediatric respiratory failure score was developed and validated against another patient subset.²⁷ This led to attempts to perform a randomized controlled trial of ECMO in pediatric respiratory failure patients using pediatric respiratory failure score guidelines. The trial was stopped early when interim analysis revealed a mortality of 18% in the control group (contrasted with a predicted mortality of 36% from earlier evaluations), and the likelihood of achieving statistical significance between ECMO and control groups was deemed futile. This trial was also likely impacted by changes that occurred in the management of respiratory failure: positive inspiratory pressure in the historical group (from which the pediatric respiratory failure score was developed) and peak inspiratory pressure in the control subjects in the randomized controlled trial were quite different (47 ± 15 vs. 34 ± 6 cm H₂O), as were the levels of $PaCO_2$ (56 ± 17 vs. 66 ± 20 mm Hg). In addition, this trial occurred at the time of rising use of high-frequency ventilation (allowed in the control group), inhaled nitric oxide, prone positioning, and surfactant, and while another clinical trial with liquid ventilation was ongoing.^{28–31} Because of the difficulties observed with this and other randomized trials of therapies in respiratory failure, especially with regards to consensus agreement on entry criteria, patient management techniques, and end points, enthusiasm for another

randomized controlled trial of ECLS in pediatric respiratory failure has been dim. Despite the absence of any randomized controlled trial, ECLS continues to be used successfully in an ever-expanding population of children.

Perhaps the most successful randomized trial of ECLS is that recently reported from the UK regarding adult respiratory failure: CESAR (Conventional versus ECMO Support in Adult Respiratory Failure).³² This trial involved adults who were randomized to conventional care or transferred to one adult ECMO center (Glenfield Hospital, Leicester, England) if appropriate inclusion criteria (based on a Murray score >3 or refractory hypercapnia with pH < 7.20) were met. Of 180 patients entered into the CESAR trial, equally divided between conventional and ECMO arms, survival at 6 months without disability was significantly higher in the ECMO arm than in the conventional arm (63% vs. 47%, $p = 0.03$); the estimated number to treat to achieve an additional life saved was six. The Data Safety and Monitoring Board stopped the trial for efficacy reasons after 180 patients had been entered. An economic analysis concluded that ECLS patients met “acceptable” cost-adjusted quality per life-year costs of £19,252, comparable to treatment of other conditions such as breast cancer.³³

Despite the impressive results, the CESAR trial continues to generate controversy, especially regarding the fact that seventeen of the ninety patients randomized to the ECLS arm did not receive ECLS. These seventeen patients were treated at the ECLS center and fourteen survived. In the minds of many clinicians, the finding that patients with severe respiratory failure had improved survival in the ECLS center, whether or not they progressed to the need for ECLS, validated the fact that clinicians in such centers provide optimal care for these patients—and that having ECLS as a tool in the algorithm of care adds a survival benefit.^{34–36} To others, this fact makes the trial results “invalid” because if the seventeen patients who did not receive ECLS are removed from the analysis, the survival benefit for ECLS is not significantly different from conventional care.

Another criticism of the CESAR trial is that patients managed in the “conventional” sites did not receive a specific algorithm of care, although use of low-tidal volume, pressure-limited ventilation was advocated. To many, the lack of mandated algorithmic care represents the “true” manner in which clinicians in non-ECLS sites provide care and makes the study even closer to what happens in “real life.” To others, the lack of mandated algorithmic care only means that perhaps overall care in the ECLS center was “better,” and the use of ECLS had nothing to do with the observed improved outcome. Also of interest (but without further analysis or explanation) is that patients in the ECLS center received corticosteroids more often than other patients. In summary, the CESAR trial provides important, current-era data and shows a positive benefit for use of ECLS in adult respiratory failure, but also raises questions for continued research and discussion.

On the heels of the publication of the CESAR trial came the H1N1 influenza epidemic of 2009–2010. Early reports of survival in patients with severe respiratory or multiple organ failure secondary to H1N1 from Australia and sites where the epidemic hit before moving west to North America revived interest among many clinicians for use of ECLS in adults. The finding that many patients succumbing to H1N1 were previously healthy young adults may also have impacted the willingness of clinicians to try something outside conventional mechanical ventilation.^{37–39} Because there are only a few centers in the United States that provide adult ECMO, many ECMO centers were overwhelmed with requests for patient transfer. An H1N1 website developed by the ELSO organization and a U.S. “bed board” [map](#) that listed open centers with contact numbers were quickly established to track patients and provide clinicians with needed expertise and assistance. To date, 263 patients have been entered into the ELSO H1N1 database with 63% survival. Two-thirds of patients are older than the age of 18 years.⁴⁰

The use of extracorporeal techniques today continues to rise in every patient age group. New equipment, better understanding of pathophysiology, improved patient management, and increased interest in ECLS in a wider variety of patients makes it an exciting time in this field.

The following focuses on the use of ECLS in, predominantly, the venoarterial mode, because the use of venovenous techniques is discussed in [Chapter 21](#).

Criteria for Extracorporeal Life Support

The ability to determine selective criteria for respiratory failure that will separate those patients who will survive with conventional care from those who will not, and thus are most in need of alternative techniques such as ECLS, is a pursuit somewhat similar to the Holy Grail. Despite the creation of multiple severity indices and subsequent evaluation of their efficacy in determining risk of death or morbidity, none have proven sustainable over time or universally correct in predicting outcome.^{41–46} Nonetheless, the most commonly used criteria for determination of respiratory failure and candidacy for ECLS are the following:

1. Alveolar-to-arterial [oxygen](#) difference ($AaDO_2$), which is calculated using the alveolar gas equation:

$$AaDO_2 = [Fi_{O_2} \times (P_B - P_{H_2O}) - Pa_{CO_2}/RQ] - Pa_{O_2}$$

where P_B represents barometric pressure (760 mm Hg at sea level), P_{H_2O} represents pressure of water vapor (47 mm Hg), and RQ is assumed to be

1. $AaDO_2$ has been historically used in neonatal respiratory failure. An $AaDO_2$ of greater than 610 over 8 hours correlated to 80% mortality in

neonatal respiratory failure in historical controls. Among pediatric patients, an AaDO₂ of greater than 470 was noted by Timmons to be 81% predictive of death, based on data published in 1991.

2. Oxygenation index (OI), which is calculated as follows:

$$OI = \frac{FI_{O_2} \times \text{mean airway pressure (cm H}_2\text{O)} \times 100}{Pa_{O_2} \text{ (mm Hg)}}$$

An OI greater than 40 predicted mortality of greater than 80% (historically). An OI of 25 to 40 predicted mortality of 50% to 80% (historically). An OI greater than 40 or remaining greater than 25 over several hours continues to be associated with high mortality in neonates, pediatric patients, and adults (especially in those following lung transplantation). In pediatric respiratory failure, high OI even at seemingly short duration of mechanical ventilation of less than 24 hours, is associated with high mortality, as shown in [Figure 20-2](#).⁴⁶

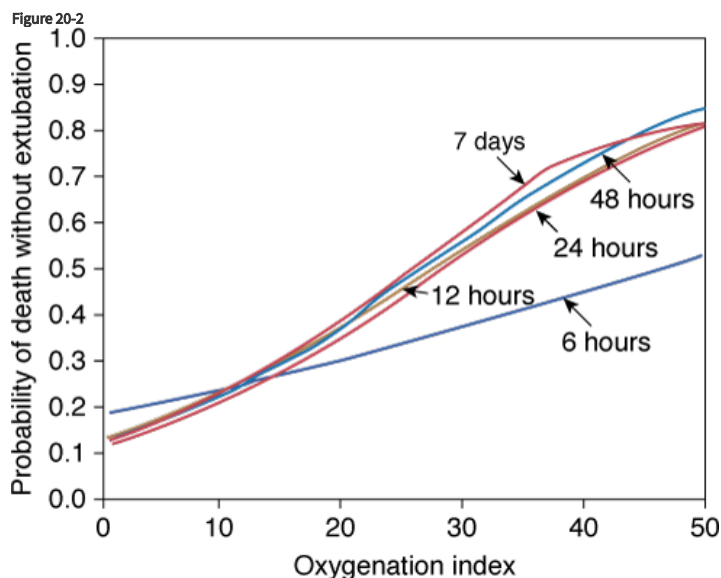
3. Compliance (C), calculated as $C = \Delta \text{volume} / \Delta \text{pressure}$, or $C = \text{tidal volume} / (\text{peak inspiratory pressure} * \text{PEEP})$, although it is preferable to use plateau pressure rather than peak inspiratory pressure. Compliance values of less than 0.5 mL/cm H₂O have been used in the selection of adult patients for ECLS.

4. Intrapulmonary shunt greater than 30% to 50% on FI_{O2} greater than 0.6. Shunt has been predominantly used as a selection criterion in adult ECLS.

5. Murray score greater than 3. The score is based on Pa_{O2}/FI_{O2}, PEEP, compliance, and number of quadrants exhibiting disease on chest radiograph. The score ranges from 0 to 4, is used in adult ECLS, and was an entry criterion in the CESAR trial.

6. Ventilatory failure. Hypercarbia with persistent pH less than 7 on high ventilator support, such as peak inspiratory pressure greater than 40 cm H₂O.

7. Pa_{O2}/FI_{O2}. Calculation is illustrated by the following example: for Pa_{O2} 50 torr, and FI_{O2} 100% (1.0), Pa_{O2}/FI_{O2} is 50/1, or 50. Pa_{O2}/FI_{O2} values of 50 to 100 torr have been described in adult ECLS.



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com
Copyright © The McGraw-Hill Companies, Inc. All rights reserved.

Oxygenation index and mortality. The presence of a high oxygenation index at any time during mechanical ventilation was associated with increased risk of death without extubation. (Adapted, with permission, from Trachsel et al.⁴⁶)

Criteria for Cardiopulmonary Failure

If it is difficult to define criteria for respiratory failure, it is equally so when ECLS is considered for cardiac or multiple organ dysfunction. Although no specific criteria have been identified and universally accepted, the following are suggestions from the literature and clinical experts:⁴⁷⁻⁵⁰

1. Plasma lactate persistently greater than 5 mM/L.
2. Mixed venous oxygen saturation ($\bar{S}VO_2$) less than 55% at an estimated cardiac index of at least 2 L/min.

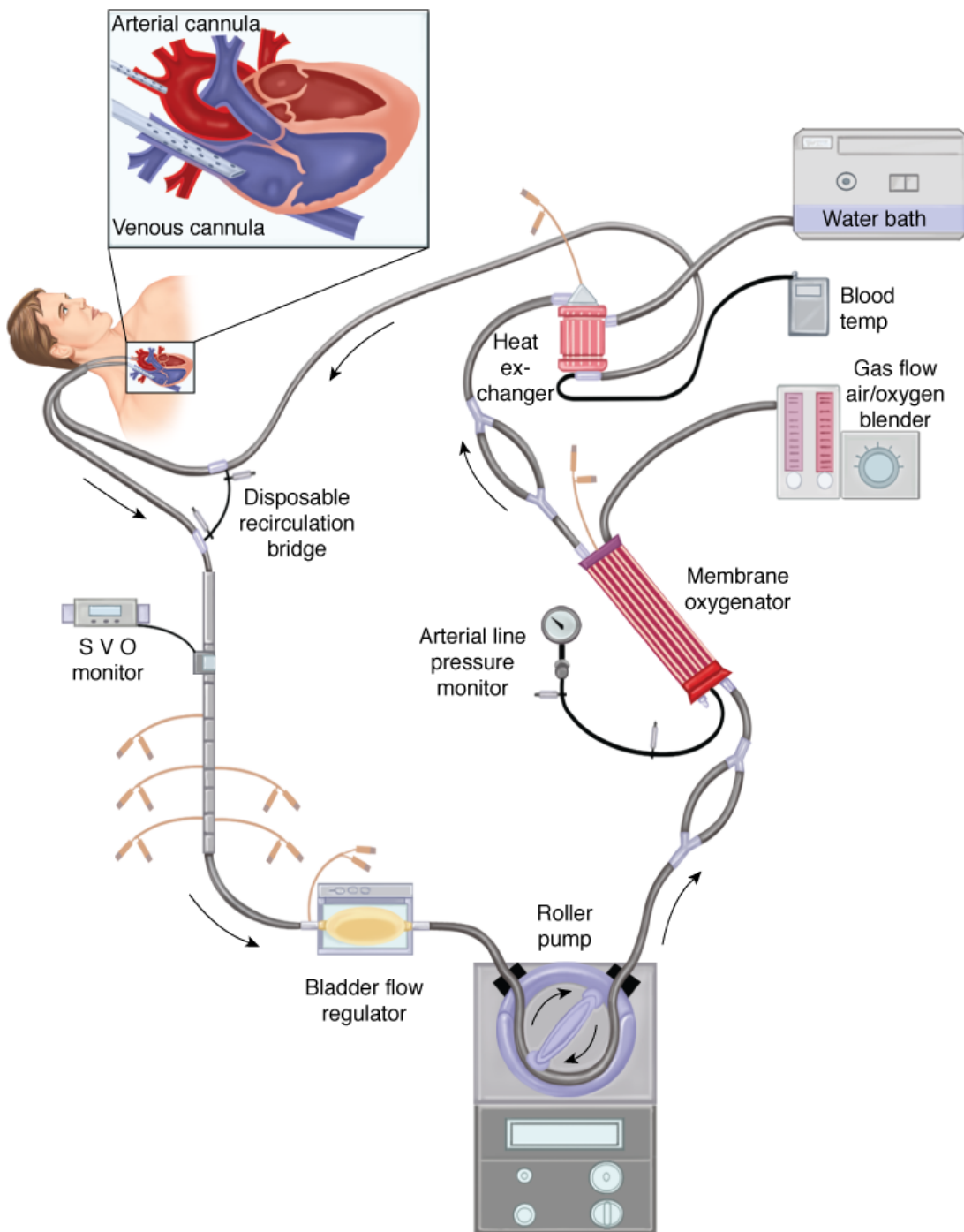
4. Intractable arrhythmia with hemodynamic compromise.
5. Cardiac arrest.
6. Inotrope score greater than 50 for 1 hour or greater than 45 for 8 hours,^{50,51} calculated as

$$\text{Dopamine (mcg/kg/min)} + \text{Dobutamine (mcg/kg/min)} + 100 \times \text{Epinephrine (mcg/kg/min)}.$$
 A modified inotrope/vasoactive score, calculated as $\text{Dopamine (}\mu\text{g/kg/min)} + \text{Dobutamine (}\mu\text{g/kg/min)} + 100 \times \text{Epinephrine (}\mu\text{g/kg/min)} + 100 \times \text{Norepinephrine (}\mu\text{g/kg/min)} + \text{Milrinone (}\mu\text{g/kg/min)} \times 10$ (15 is also used by some clinicians) $+ 10,000 \times \text{vasopressin dose (}\mu\text{g/kg/min)}.$
7. Failure to wean from cardiopulmonary bypass.

Venoarterial Cannulation Modes

Venoarterial ECLS has been the predominant mode of support in infants and children for many years secondary to lack of adequately sized venous vessels to obtain the amount of blood flow needed to support the patient. [Figure 20-3](#) is an example of a typical venoarterial ECLS circuit with cervical cannulation, a roller head pump, and silicone membrane lung. [Figure 20-4](#) depicts more current equipment with centrifugal pump and hollow-fiber oxygenator. With the advent of double-lumen, single cannulas, and improvements in flow characteristics of available cannulas, venovenous support is now becoming more popular and feasible even among neonates.^{51,52} Gone are the days when modifications of thoracostomy or endotracheal tubes provided the “cannulas” for ECLS support; wire-reinforced, thin-walled cannulas now permit excellent flows at smaller internal and external diameters.⁵³ Venoarterial support, however, remains a mainstay of support in patients with combined cardiopulmonary dysfunction or in those with primary cardiac failure. [Figures 20-5](#) and [20-6](#) show the changes in mode of cannulation by category and diagnostic group.

Figure 20-3



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com

Copyright © The McGraw-Hill Companies, Inc. All rights reserved.

Venoarterial ECLS circuit with roller-head pump and silicone membrane oxygenator. Note venous saturation device on drainage limb; bladder reservoir device is before pump head. Silicone membrane oxygenator has now been almost totally replaced by new hollow-fiber devices. Heat exchanger device rewarms or cools blood before its return to patient.

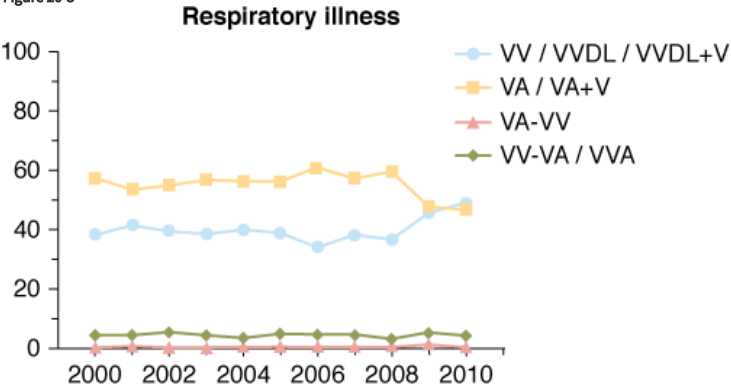
Figure 20-4



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com
Copyright © The McGraw-Hill Companies, Inc. All rights reserved.

Centrifugal pump and hollow-fiber oxygenator system. These systems often have lower priming volumes, are easier and faster to set up, and do not require gravity drainage. Note that the oxygenator and pump head should be below the level of the patient.

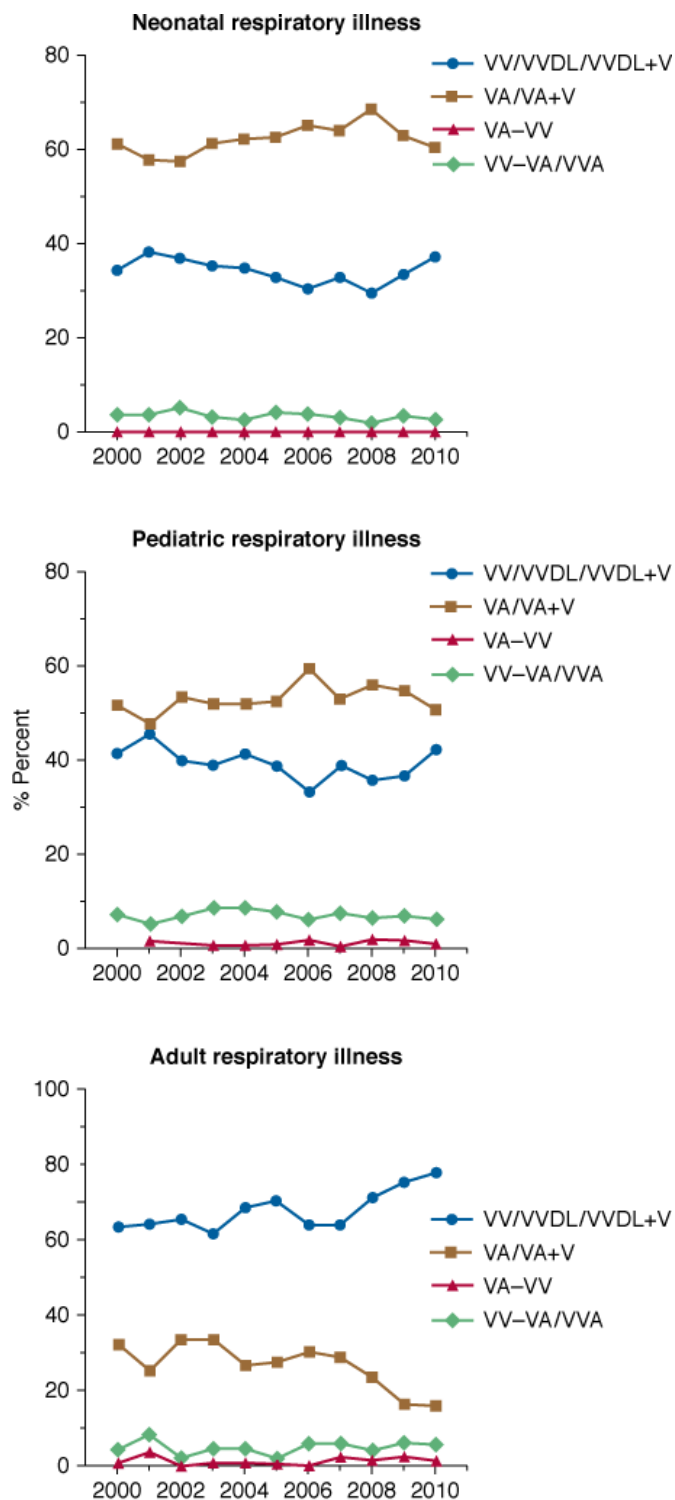
Figure 20-5



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com
Copyright © The McGraw-Hill Companies, Inc. All rights reserved.

Cannulation mode for patients with respiratory failure (all ages). **VA**, venoarterial; **VA+V**, additional venous cannula added; **VA-VV**, venoarterial cannulation changed to venovenous during ECLS course; **VV**, venovenous; **VVDL**, venovenous double lumen; **VV-VA**, venovenous cannulation changed to venoarterial during ECLS course

Figure 20-6



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com
Copyright © The McGraw-Hill Companies, Inc. All rights reserved.

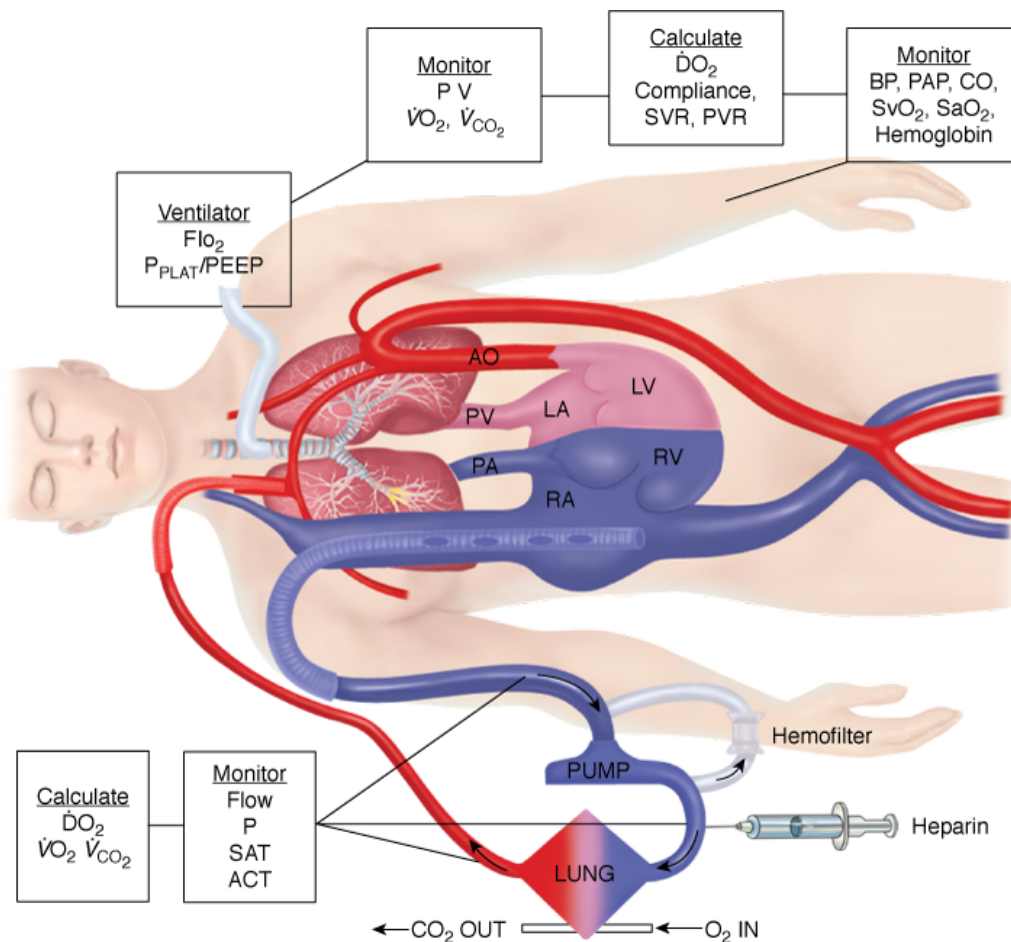
Cannulation mode by age. Note increase in venovenous support over past several years. VA, venoarterial; VA+V, additional venous cannula added; VA-VV, venoarterial cannulation changed to venovenous during ECLS course; VV, venovenous; VVDL, venovenous double lumen; VV-VA, venovenous cannulation changed to venoarterial during ECLS course.

Cervical Cannulation

In neonates, use of the internal jugular vein and right common carotid artery is the predominant venoarterial cannulation technique. Figure 20-7 is an example of cannulation through the internal jugular vein and carotid artery for venoarterial extracorporeal support with monitoring parameters that can be used for patient management. While centers may vary somewhat in cervical cannulation technique, most use an open approach to vessel access.⁵⁴ Repair of the carotid artery at decannulation is controversial, practiced routinely in some centers and avoided in others.⁵⁵⁻⁵⁸

Loading [Contrib]/a11y/accessibility-menu.js

Figure 20-7



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com
Copyright © The McGraw-Hill Companies, Inc. All rights reserved.

Venoarterial cannulation. Boxes represent calculations and parameters that can be measured by monitoring site. ACT , activated clotting time; BP , arterial blood pressure; CO , cardiac output (estimated); $\dot{D}O_2$, oxygen delivery; P , pressure; PAP , pulmonary artery pressure (only in patients with a pulmonary artery catheter in place); $PEEP$, positive end-expiratory pressure; P_{PLAT} , plateau pressure; PVR , pulmonary vascular resistance; SaO_2 , arterial oxygen saturation; Sat , saturation of blood; $s\bar{V}O_2$, venous oxygen saturation; SVR , systemic vascular resistance; V , volume; $\dot{V}CO_2$, carbon dioxide; $\dot{V}O_2$, oxygen consumption. (Courtesy of Robert Bartlett, MD, with permission.)

Follow-up studies of repaired vessels has noted patency in 80% to 100% of repaired vessels, but stenosis in repaired vessels also has been noted in several studies, as has the development of aneurysms or pseudoaneurysms of the vessel with need for emergent surgical intervention. In one follow-up report of neurodevelopmental outcome from pediatric patients receiving support for cardiac dysfunction, use of the carotid artery for vascular access was not associated with neurologic dysfunction. Similarly, small comparative studies of neonates, with or without carotid repair at decannulation, did not find neurodevelopmental changes at follow-up between groups and noted good collateral flow in patients where the carotid had been ligated. Despite the lack of evidence in favor of repair of the carotid artery, there is increasing interest, especially in older patients, in avoiding the carotid artery for cannulation because the potential risk for stroke seems higher (anecdotally) despite no proven evidence of this concern. Only long-term outcome studies, up to the ages when stroke becomes more common, will be able to definitely answer whether carotid artery ligation or repair following ECLS is an associated risk factor.

Internal jugular cannulation can be performed with a semi-Seldinger technique or via direct venotomy and insertion of the cannula. Some centers also insert a small retrograde venous drainage cannula to the level of the jugular bulb to obtain more venous drainage from the cerebral circuit and/or to monitor cerebral venous saturation as an indication of adequacy of perfusion to the brain and oxygen delivery and extraction (this is presumably even more important in venovenous support).^{59,60} Some clinicians report having been able to increase venous drainage by up to 30% in neonates with this technique, although others have noted clotting difficulties in the retrograde cannula and have abandoned this technique. The effects of ligation of the internal jugular vein on cerebral venous drainage and risk for intracranial hemorrhage from venous stasis or hypertension has not been completely defined. Repair of the vein at decannulation is not practiced routinely, although patency has been described. For cardiac patients in whom future surgeries may involve use of the superior venous system, such as single ventricle patients, judicious use of the internal jugular vein for ECLS and consideration for repair if cannulation occurs should be discussed.

Loading [Contrib]/a11y/accessibility-menu.js

Central Cannulation

ECLS can also be provided through direct insertion of large-bore cannulas into the right atrium and into the aortic arch. This method of cannulation allows the largest volume of blood to be obtained for support. This technique is used predominantly in patients transitioned to ECLS directly from cardiopulmonary bypass or following acute deterioration (including arrest) in the early postoperative period when the sternum can be easily reopened.^{61,62} It has also recently been identified as a potential factor in successful support of patients with septic shock and multiple organ failure, who may require high levels of flow to obtain adequate oxygen delivery. In the most cited report, children in septic shock who received central cannulation were unexpectedly found to have improved survival as compared to those in whom other cannulation strategies were used (73% vs. 38% survival, $p = 0.05$, $n = 11$).⁶³ These patients required sternotomy for cannula insertion after peripheral cannulation proved inadequate for support or vascular access could not be obtained. The major differentiating feature between the groups was the higher flow rates obtained with the central cannulation mode. The authors speculated that this improved flow provided better oxygen delivery and resolution of organ failure with resultant improved survival. A follow-up report of twenty-three children with septic shock supported with ECLS over a 9-year period using central cannulation noted 74% survival to discharge.⁶⁴ Although central cannulation in patients without a prior sternotomy is not routine in most centers, the aforementioned reports have led it to be considered in non-cardiac-surgery patients and we have used it successfully in septic patients in whom peripheral cannulation was ineffective. Bleeding from the mediastinal site continues to be a major complication.

Femoral Cannulation

In children older than the age of 2 years or roughly weighing more than 15 kg, femoral vessels may be adequate for venoarterial ECLS. Concern with use of the carotid artery in older patients has already been discussed. Thus, in larger patients (especially adults), femoral cannulation is often preferred.

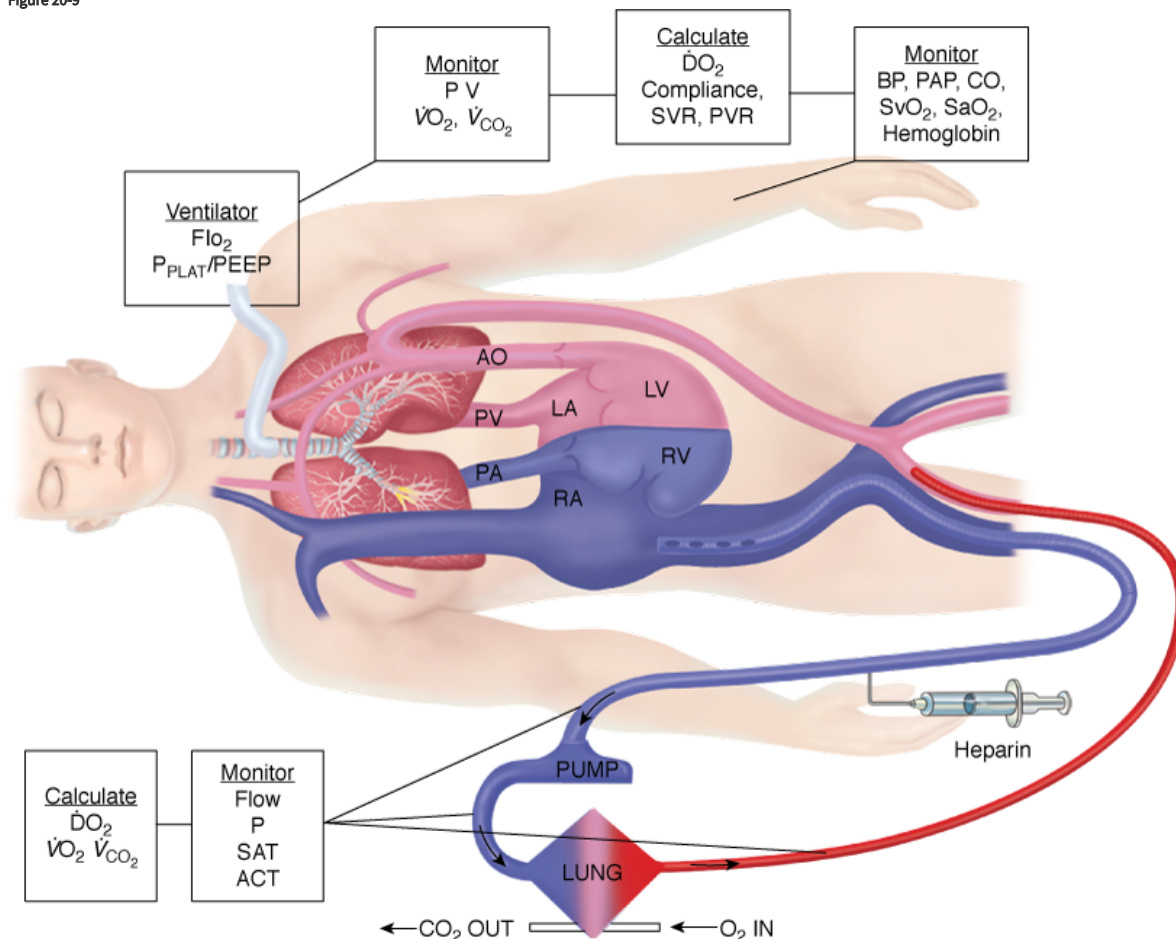
Figure 20-8 is a typical example of femoral/arterial cannulation. Venous cannulation can be performed percutaneously or with open venotomy.^{65–67} Optimally, a long, large-bore cannula is passed up the femoral vein to the junction of the inferior vena cava-right atrium. Figure 20-9 depicts a typical femoral venoarterial cannulation. It should be noted that as the flow characteristics of a cannula are dependent on internal diameter and length, longer cannulas will have higher resistance to flow; however, achieving access near the right atrium allows for the largest pool of venous blood to be obtained, and, thus, short femoral cannulas (while having less resistance) will frequently not result in adequate venous drainage for ECLS support, especially in small patients. Femoral arterial access can be performed by percutaneous or open cannulation. Typically, a short (18 to 25 cm) cannula is used for ECLS return. In access via the femoral vessels, integrity of distal perfusion or venous drainage of the limb is at risk. Measuring vessel size with bedside ultrasound and picking a cannula with an internal diameter slightly less than the vessel can help allow flow around the cannula and prevent blood flow inadequacies to the distal limb.

Figure 20-8



Modified femoral venoarterial ECMO: For patients usually >15 kg, adequate support can be obtained from femoral venous and femoral arterial cannulation. In this depiction, the patient also has sheaths for cardiac catheterization in the right groin. (Courtesy of Heidi Dalton MD, with permission.)

Figure 20-9



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com
 Copyright © The McGraw-Hill Companies, Inc. All rights reserved.

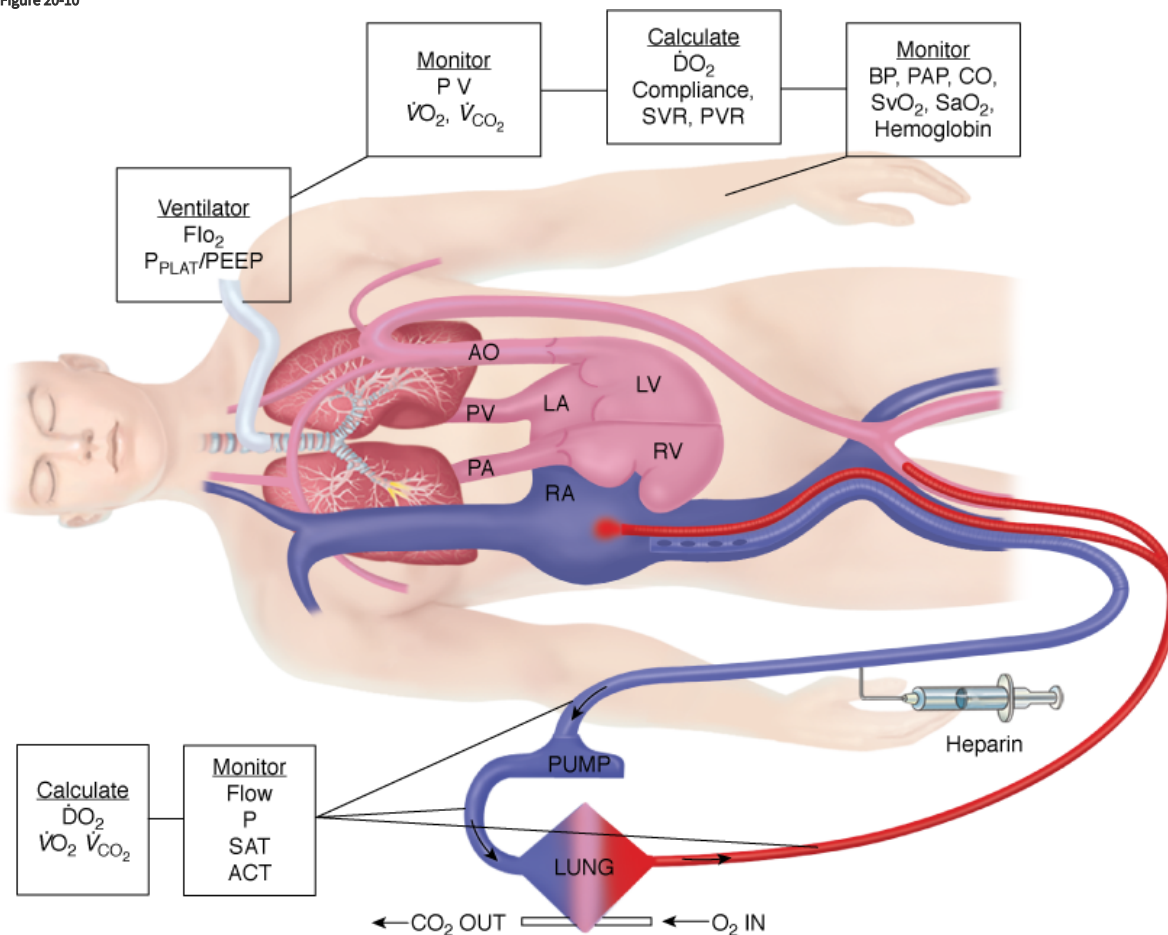
Femoral venoarterial cannulation. Long femoral drainage cannula to right atrium and short arterial return cannula in femoral artery. Boxes represent calculations and parameters that can be measured by monitoring site. ACT, activated clotting time; BP, arterial blood pressure; CO, cardiac output (estimated); DO_2 , oxygen delivery; P, pressure; PAP, pulmonary artery pressure (only in patients with a pulmonary artery catheter in place); PEEP, positive end-expiratory pressure; P_{PLAT} , plateau pressure; PVR, pulmonary vascular resistance; SaO_2 , arterial oxygen saturation; Sat, saturation of blood; SvO_2 , venous oxygen saturation; SVR, systemic vascular resistance; V, volume; $\dot{V}\text{CO}_2$, carbon dioxide; $\dot{V}\text{O}_2$, oxygen consumption. (Courtesy of Robert Bartlett, MD, with permission.)

Placement of small drainage cannulas (for venous engorgement) or distal perfusion cannulas (for distal arterial ischemia) have both been described and used efficiently to prevent compartment syndrome or limb loss.⁶⁸ Despite these maneuvers, however, limb injury to the extent of amputation has been described several times following femoral access. Thus, close neurovascular monitoring is required.

As venous flow obtained from the femoral route is usually less than that obtained from the internal jugular or central venous sites, the amount of support provided in this mode of venoarterial ECLS is often less than can be obtained with other cannulation techniques. This results in a lower arterial oxygen saturation (SaO_2) secondary to increased mixing of desaturated blood from the patient (assuming the patient has severe respiratory failure and impaired gas exchange) than with other forms of venoarterial cannulation. Oxygen saturation often will be 80% instead of 90% to 100% and will mimic that seen with venovenous support. Another important factor with femoral venoarterial support is that the oxygenated arterial return from the ECLS circuit runs retrograde up the aorta to flow coming out of the native heart. How far up the aorta the oxygenated ECLS blood extends is a product of the amount of bypass performed and the amount being ejected from the left ventricle. One concern with femoral venoarterial bypass is that with severe respiratory failure, blood from the left ventricle will be about the same saturation as that in the right atrium, and this relatively desaturated blood will be what is perfusing the upper body, especially the head and heart. This can result in a “blue upper body and red lower body” phenomenon. Whether this is harmful depends on the extent of desaturation and patient-specific factors. For a patient with evidence of myocardial ischemia, having desaturated blood flowing into the coronary vessels may not be optimal. Similarly, if the patient is exhibiting signs of inadequate cerebral oxygenation (Loading [Contrib]/a11y/accessibility-menu.js) oscopy monitoring or some other parameter), then this may be an indication that oxygenation is inadequate. If

upper-body [oxygen](#) delivery is deemed to be significantly impaired, one technique is to place a venous cannula into the right internal jugular vein or in the femoral vein and direct some of the oxygenated ECLS return into this cannula by means of a Y connector on the return limb of the circuit. [Figure 20-10](#) illustrates this method of hybrid femoral venoarterial cannulation with an additional venous cannula providing some oxygenated return. Although the figure depicts two cannulas in one femoral vein, this approach is not common; most clinicians either will place the additional venous cannula in the right internal jugular vein or in the opposite femoral vein. This increases the venous saturation of blood in the right atrium and thus the saturation of blood being ejected out the left ventricle. Careful monitoring of flow in both the venous cannula and the femoral arterial cannula should be performed to prevent one cannula from receiving inadequate flow (as blood will always take the path of least resistance) and clotting. Manipulation of flows to multiple cannulas “Y-ed” into the circuit can be done with a simple screw clamp or other devices. The need for increased venous drainage in femoral venoarterial ECLS can also be achieved by adding another venous cannula (usually in the right internal jugular) and “Y-ing” it into the venous drainage side of the circuit.

Figure 20-10



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com

Copyright © The McGraw-Hill Companies, Inc. All rights reserved.

Hybrid femoral venoarterial cannulation with additional femoral cannula for additional oxygenation to venous circulation. Although the depiction shows additional cannula in same site as venous drainage cannula, it is more common to place additional cannula in the opposite femoral vein or internal jugular vein. Boxes represent calculations and parameters that can be measured by monitoring site. *ACT*, activated clotting time; *BP*, arterial blood pressure; *CO*, cardiac output (estimated); *DO₂*, oxygen delivery; *P*, pressure; *PAP*, pulmonary artery pressure (only in patients with a pulmonary artery catheter in place); *PEEP*, positive end-expiratory pressure; *P_{PLAT}*, plateau pressure; *PVR*, pulmonary vascular resistance; *SaO₂*, arterial oxygen saturation; *Sat*, saturation of blood; *S $\bar{V}$ O₂*, venous oxygen saturation; *SVR*, systemic vascular resistance; *V_i*, volume; $\dot{V}_{CO_2}$, carbon dioxide; $\dot{V}O_2$, oxygen consumption. (Courtesy of Robert Bartlett, MD, with permission.)

Although other vessels, such as the axillary and subclavian, have been used as access for venoarterial ECLS, reports of their use are infrequent. Improved cannula designs with better flow rates and characteristics, may allow improved access to other vessels, even to the point of using umbilical vessels for premature infant support, in the future. Careful attention to neurovascular monitoring is paramount when vessels, especially outside the norm, are cannulated.⁶⁹

Physiology of Venoarterial Extracorporeal Life Support

Loading [Contrib]/a11y/accessibility-menu.js

Venoarterial ECLS allows for drainage from the right side of the heart, thus reducing blood flow through the native cardiopulmonary circuit. In patients with impaired pulmonary gas exchange, diversion of venous blood into the ECLS circuit, where it undergoes oxygenation and removal of carbon dioxide, and direct return of oxygenated blood into the arterial system allows for increased systemic oxygenation. The more blood diverted into the ECLS circuit, the more bypass is performed and the higher the systemic oxygen delivery provided by ECLS. Increasing ECLS flow will decrease the proportion of blood flowing through the diseased lungs, which is ejected out the left ventricle to mix with the oxygenated return from the ECLS circuit. Thus, at higher ECLS flows, systemic oxygen delivery and SaO_2 will increase. This has led to the adage, “If the O’s are low, turn up the flow.” Similarly, at a constant ECLS flow without manipulations in oxygenation of blood within the membrane oxygenator, a simple herald of improving native respiratory function is an increase in SaO_2 , as the patient’s lungs become more efficient at gas exchange and the arterial oxygen content (CaO_2) of ejected left-ventricular blood increases. Just as in normal care of a critically ill patient, understanding the role of CaO_2 and oxygen delivery in ECLS is important. As a reminder, CaO_2 is calculated as: $(\text{hemoglobin g/dL} \times \% \text{SaO}_2 \times 1.36 \text{ mL/g}) + (0.003 \text{ mL/dL} \times \text{PaO}_2) \times 10$. Thus, the major contributors to CaO_2 are hemoglobin concentration and SaO_2 . PaO_2 is of little impact on CaO_2 unless severe anemia exists. Figure 20-7 shows a typical venoarterial circuit with parameters that can be followed or calculated.

To determine the overall oxygen delivered to the patient during ECLS, both the CaO_2 and flow must be considered. The following example illustrates the principles involved:

Assume that perfusate blood is 100% saturated with a PaO_2 of 500 torr, as might be seen with 100% FiO_2 applied to the membrane lung. If native respiratory failure is so severe that no oxygenation occurs in the pulmonary circuit, blood returning from the native lungs and being ejected out the left ventricle will have the same saturation as right-atrial venous blood. With normal oxygen delivery and extraction, right-atrial venous saturation is 75% and the partial pressure of oxygen (P_{O_2}) 35 torr. At a hemoglobin of 15 g/dL, the perfusate oxygen content is 22 mL/dL and the right- and left-atrial ventricular oxygen content is 15 mL/dL. If 50% of the native blood flow is diverted into the ECLS circuit and 50% flows through the native cardiopulmonary circuit, the measured systemic oxygen content in the patient will be a combination of these amounts and will result in an oxygen content of 18.5 mL/dL, which corresponds to an SaO_2 of 90% and a PaO_2 of 55 torr. These are the values that will be obtained from the patient’s arterial blood, assuming that the blood is being measured past the site where the ECLS return enters the patient so that mixing with native output has occurred.

Thus, an improvement in oxygenation during venoarterial ECLS indicates either (a) improving lung function, (b) decreasing native cardiac output (at constant ECLS flow), or (c) increasing ECLS flow (if native cardiac output is constant). The amount of oxygenation that occurs in the membrane lung, is dependent on gas exchange properties, thickness of the blood film, hemoglobin concentration, the residence time of red blood cells within the membrane lung, and the gradient between inlet venous blood and the oxygen content in the ventilating gas flow.^{70,71} Assume that the mixed venous oxygen tension (PvO_2) reaching the membrane lung is 35 to 40 torr and even with room air, the P_{O_2} of ventilating gas is around 100 torr with a saturation of 100%, oxygen will flow into the venous blood and saturation at the outlet of the oxygenator will be optimally 100%. Increasing the FiO_2 of the sweep gas may increase the P_{O_2} in outlet blood but will not increase oxygen saturation to greater than 100%; thus, the increase in oxygen content at higher sweep gas FiO_2 will be minor unless the patient is severely anemic. If the patient requires more oxygen delivery and blood exiting the membrane lung is 100% saturated, then increasing ECLS flow is needed. For many oxygenators, adequate saturation can be achieved with only room air in the sweep gas. As membrane lung efficiency may decrease over time, an increase in FiO_2 of the sweep gas may be required to achieve 100% outlet saturation. Each oxygenator is rated for optimal blood flow (defined as the rate of venous blood flow which results in an increase in SaO_2 from 75% to 95% as blood travels through the device from inlet to outlet). Increasing blood flow rates above the “rated flow” impairs optimal oxygen delivery by the device.

CO_2 removal with venoarterial ECLS is extremely efficient and is dependent on membrane lung surface area, material, geometry, and membrane lung ventilating gas flow (usually termed the “sweep gas flow”), as well as P_{CO_2} content in the patient’s blood. Normally, sweep gas flow contains no CO_2 so the gradient for CO_2 exchange across the membrane lung is dependent on the difference between the venous blood reaching the oxygenator and the sweep gas flow. At high sweep gas flows, or with larger membrane surface areas, more CO_2 is removed from the blood, similar to increasing minute ventilation as a means of increasing CO_2 elimination in a normal state. In small patients, CO_2 elimination can be so efficient that the P_{CO_2} returning to the patient is very low and respiratory alkalosis severe. For patients cannulated via the cervical or central arterial route into the arch of the aorta, the ECLS return often reaches the left carotid artery (and thus the brain) first. Because a very low P_{CO_2} in this blood may adversely affect cerebral vascular blood flow, maintaining normal Pa_{CO_2} in the blood returning to the patient is recommended. Some centers regulate P_{CO_2} returning to the patient by manipulating sweep gas flow as a lower sweep will decrease CO_2 removal. Others blend CO_2 gas into the sweep gas mixture to decrease the gradient

across the membrane lung between native blood and sweep gas. It should be remembered that CO₂ elimination is better than [oxygen](#) uptake across both the silicone membrane and microporous/hollow-fiber oxygenator devices.^{72–75}

One important note with venoarterial support is that arterial flow coming from the ECLS circuit *increases* afterload to the left ventricle. This is of paramount importance in situations of left-ventricular dysfunction, because this increase in afterload may result in sudden and complete failure of the left heart. The majority of coronary artery perfusion during venoarterial ECLS has been shown to result from native left-ventricular ejection.⁷⁶ Given that venoarterial ECLS never provides complete bypass, because flow from bronchial and thebesian vessels will continue to the left heart, inability of the left heart to eject can result in impairment of left-ventricular myocardial flow from elevated intracardiac pressure, left-atrial hypertension, pulmonary venous hypertension, and pulmonary hemorrhage. Although a short period of cessation of left-ventricular function can often be tolerated, attempts at afterload reduction with medications such as milrinone or nitroprusside can also be helpful. In patients with prolonged left-ventricular failure, poor aortic ejection, evidence of pulmonary congestion (such as pulmonary edema or hemorrhage), an atrial septostomy can be performed to allow left-atrial blood to be drained via the right-atrial cannula in order to decrease left-atrial pressure.^{77,78} In patients with sternotomy, a left-atrial vent can also be placed and “Y-ed” into the venous side of the ECLS circuit to offload the left heart. If the patient’s cardiopulmonary function is being “tested” while clamped off ECLS before complete removal of the cannula’s to see if adequate recovery has occurred (this process is called “trialing off ECLS”), the left-atrial cannula must be removed or clamped during this process to allow normal filling of the left heart. Echocardiography can be invaluable to follow the extent of left-ventricular dysfunction, overall cardiac function and recovery, adequacy and optimal placement of ECLS cannulas, and assessment for pericardial effusion during ECLS.

Unlike cardiopulmonary bypass, ECLS is intended to provide partial (not *total*) bypass, as animal studies show that complete bypass of the pulmonary circuit is associated with ischemia.⁷⁹ The “total” bypass used during cardiac surgery is also thought to be a factor in the reperfusion injury noted in the lung, cytokine generation, and the inflammatory response observed in the post-bypass period. Thus, goals for venous saturation, arterial oxygenation, and hemodynamic support are set for each patient (usually on a daily basis) and ECLS flow adjusted to maintain set parameters. For patients with severely impaired gas exchange, the level of bypass support is set at providing adequate [oxygen](#) delivery to support tissue function. In general terms, this often equates to a starting ECLS flow of 100 to 150 mL/kg in neonates, 80 to 100 mL/kg in children, and 50 to 60 mL/kg in adults. The adequacy of support is monitored by establishing normal venous [oxygen](#) saturations (>65% to 70%), normalization of lactate levels and base deficit as well as adjunct measures such as clinical exam, vital signs such as blood pressure, neurologic alertness, urine output, near-infrared spectroscopy monitoring, and other measures used by clinicians to identify appropriate organ perfusion.⁸⁰ Flow returning from the ECLS circuit is mainly nonpulsatile. One indication of the amount of bypass being performed in a patient with adequate cardiac function is merely following the pulse pressure and systolic pulse contour on the arterial waveform. More bypass will decrease the systolic upswing and give a more narrow pulse pressure.

Several practical points regarding ECLS use in patients with predominant cardiac failure are to maintain intracardiac filling pressures at low levels so as to promote endocardial blood flow and myocardial perfusion. Thus, close attention to central venous pressure and left-atrial pressure (if available) is required. Use of echocardiography to assess cardiac function (how “full” or “empty” the heart is) and aortic outflow is very helpful during ECLS, especially in the cardiac patient and during weaning.⁸¹

Patient Populations Treated with Venoarterial Extracorporeal Life Support

[Table 20-1](#) reviews the ELSO registry for patient populations in neonatal (<30 days), pediatric (31 days to 18 years), and adult (>18 years) age groups in terms of major diagnostic categories and outcome. [Table 20-2](#) outlines major diagnostic groups within categories and outcome in the most recent time period of 2000 to 2010. [Figures 20-11](#) and [20-12](#) show the changes in ventilator settings and blood-gas parameters in age groups and respiratory failure patients. Note that pH has declined and PaCO₂ has increased over time, perhaps indicating adoption of the “permissive hypercapnia” approach to mechanical ventilation. This has not been accompanied, however, by a large change in OI over time, as might be expected if “gentler ventilation” with lower mean airway pressures was occurring. Potential explanations for the lack of change in OI is that mean airway pressure with the high frequency oscillator is often higher (at least initially) than with conventional mechanical ventilation or that clinicians are accepting greater levels of hypoxia as part of “gentler ventilation.”

Table 20-1: Patient Populations Treated with Extracorporeal Life Support and Outcome

ECLS Registry Report International Summary July, 2011				Extracorporeal Life Support Organization 2800 Plymouth Road Building 300, Room 303 Ann Arbor, MI 48109	
Overall outcomes					
	<i>Total Patients</i>	<i>Survived ECLS</i>		<i>Survived to DC or Transfer</i>	
Neonatal					
Respiratory	24,770	20,951	85%	18,558	75%
Cardiac	4,375	2,649	61%	1,723	39%
ECPR	694	438	63%	270	39%
Pediatric					
Respiratory	5,009	3,251	65%	2,785	56%
Cardiac	5,423	3,468	64%	2,609	48%
ECPR	1,347	720	53%	539	40%
Adult					
Respiratory	2,620	1,655	63%	1,428	55%
Cardiac	1,680	894	53%	660	39%
ECPR	591	225	38%	173	29%
Total	46,509	34,251	74%	28,745	62%

DC, hospital discharge; ECLS, extracorporeal life support; ECPR, extracorporeal life support during cardiopulmonary resuscitation.

Survival is off ECLS and then to hospital discharge.

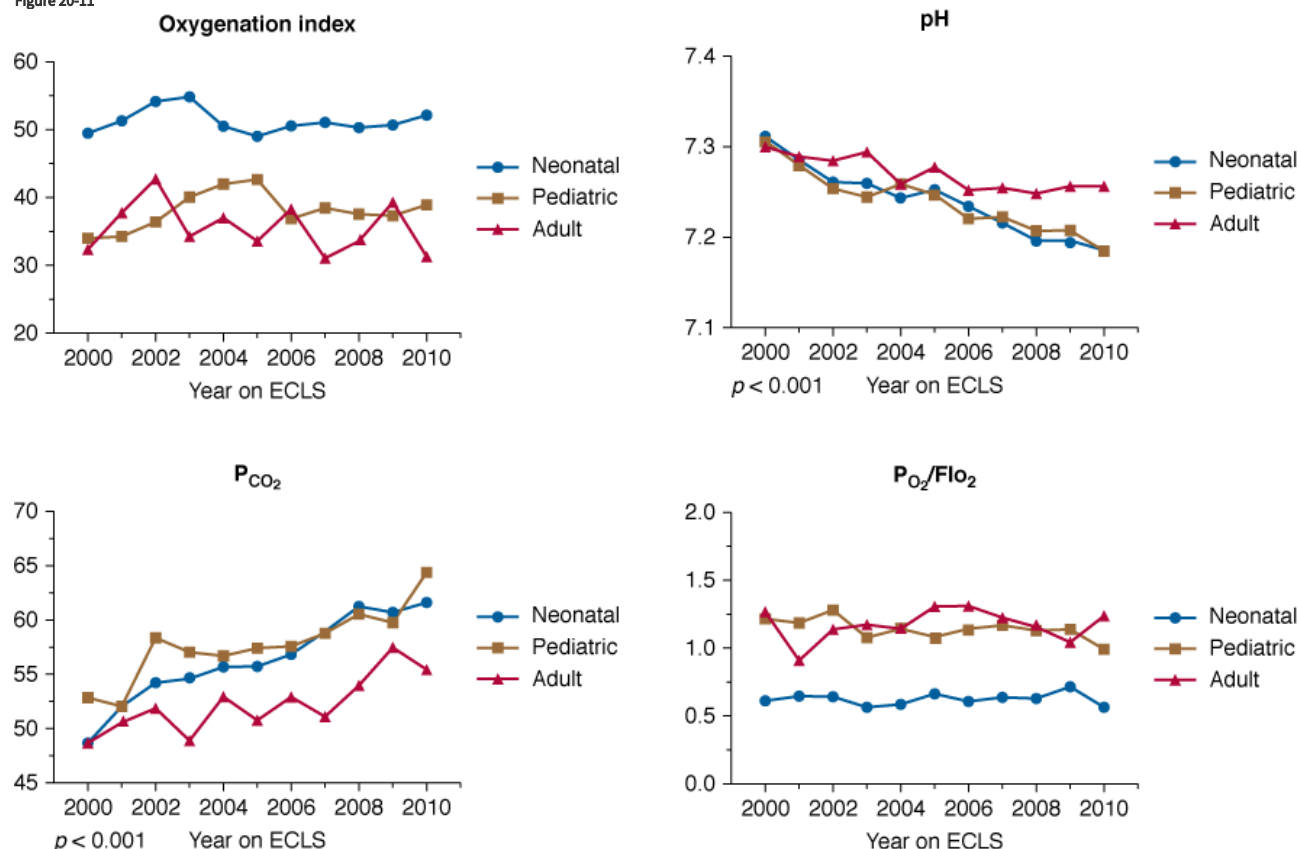
From the International Registry of the Extracorporeal Life Support Organization, Ann Arbor, MI, with permission.

Table 20-2: Outcome According to Major Diagnostic Group (2000 to 2010)

Primary Diagnosis	n (%)	Survival
Neonatal	9086	68.6
MAS	2239 (24.6)	92.9
RDS	203 (2.2)	85.2
PPHN/PFC	1820 (20.0)	75.9
Air leak syndrome	16 (0.2)	75.0
Sepsis	252 (2.8)	70.2
Other	1785 (19.7)	65.5
CDH	2767 (30.5)	46.7
Pneumonia	4 (0.04)	25.0
Pediatric	2992	55.7
Aspiration	13 (0.43)	69.2
Bacterial	134 (4.48)	62.7
ARDS, post-op/trauma	80 (2.7)	60.0
Acute respiratory failure, not ARDS	306 (10.2)	56.5
Pneumocystitis	16 (0.53)	56.3
Other	2246 (75.1)	55.1
ARDS, not post-op/trauma	185 (6.2)	54.1
Viral	12 (0.4)	50.0
Adult	1921	56.0
Viral	35 (1.8)	74.3
Aspiration	7 (0.4)	71.4
Bacterial	156 (8.1)	62.8
ARDS, post-op/trauma	143 (7.4)	60.1
Acute respiratory failure, non-ARDS	102 (5.3)	57.8
Other	1266 (65.9)	55.4
ARDS, not post-op	212 (11.0)	47.2

ARDS, acute respiratory distress syndrome; CDH, congenital diaphragmatic hernia; MAS, meconium aspiration syndrome; post-op, postoperative; PPHN/PFC, persistent pulmonary hypertension of the newborn; RDS, respiratory distress syndrome.

Figure 20-11

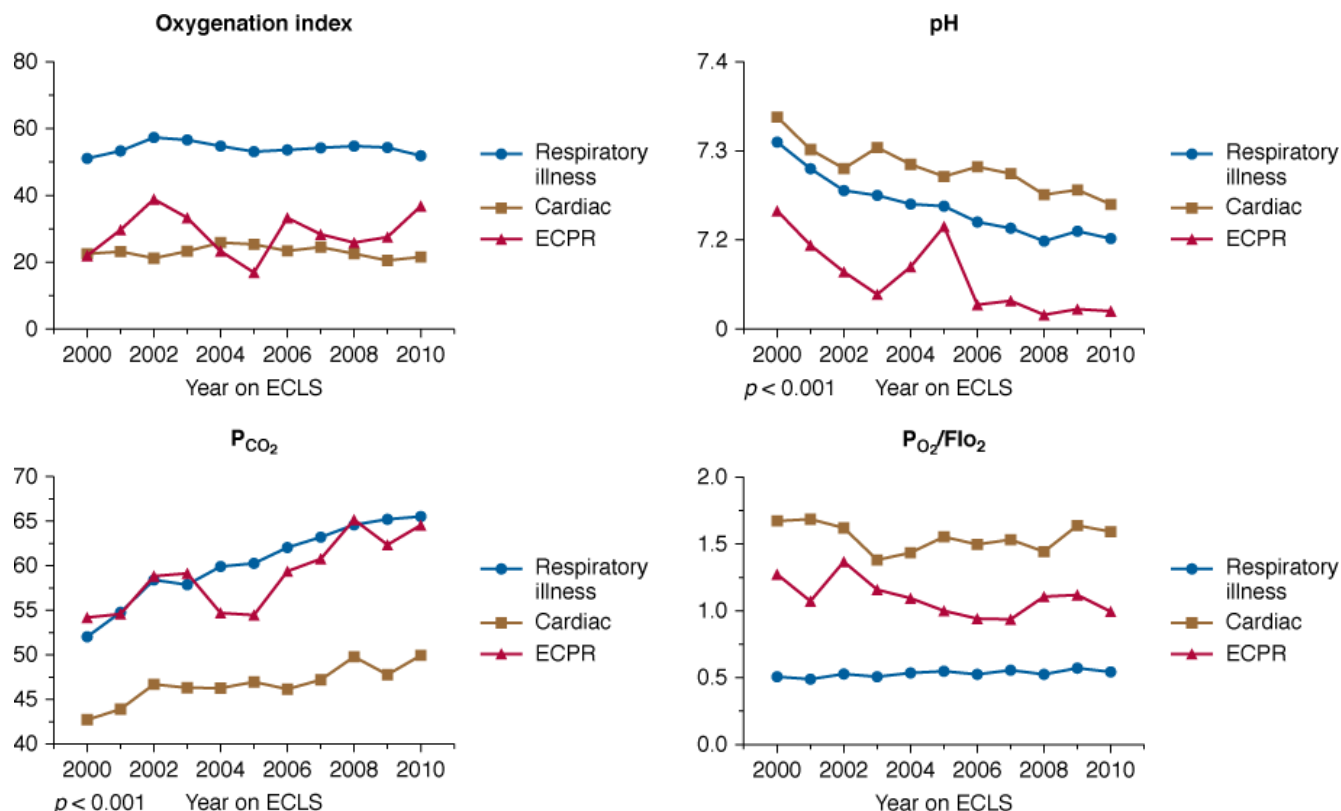


Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com

Copyright © The McGraw-Hill Companies, Inc. All rights reserved.

Severity indices and blood-gas parameters over time by age group (2000 to 2010). Oxygenation index and P_{O_2}/F_{IO_2} have decreased slightly in adults over time; pH has decreased and P_{aCO_2} has increased, possibly reflecting greater use of “lung-protective” ventilation. Increased use of high-frequency oscillation ventilation (HFOV) in neonates and pediatric patients versus adults may influence the oxygenation index because mean airway pressure is often higher with HFOV.

Figure 20-12



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com
Copyright © The McGraw-Hill Companies, Inc. All rights reserved.

Severity indices and blood-gas parameters over time by diagnostic category (2000 to 2010). Oxygenation index and $P_{O_2}/F_{I_{O_2}}$ have decreased slightly in adults over time; pH has decreased and P_{aCO_2} has increased, possibly reflecting greater use of “lung protective” ventilation. Increased use of high-frequency oscillation ventilation (HFOV) in neonates and pediatric patients versus adults may influence oxygenation index because mean airway pressure is often higher with HFOV.

Neonatal

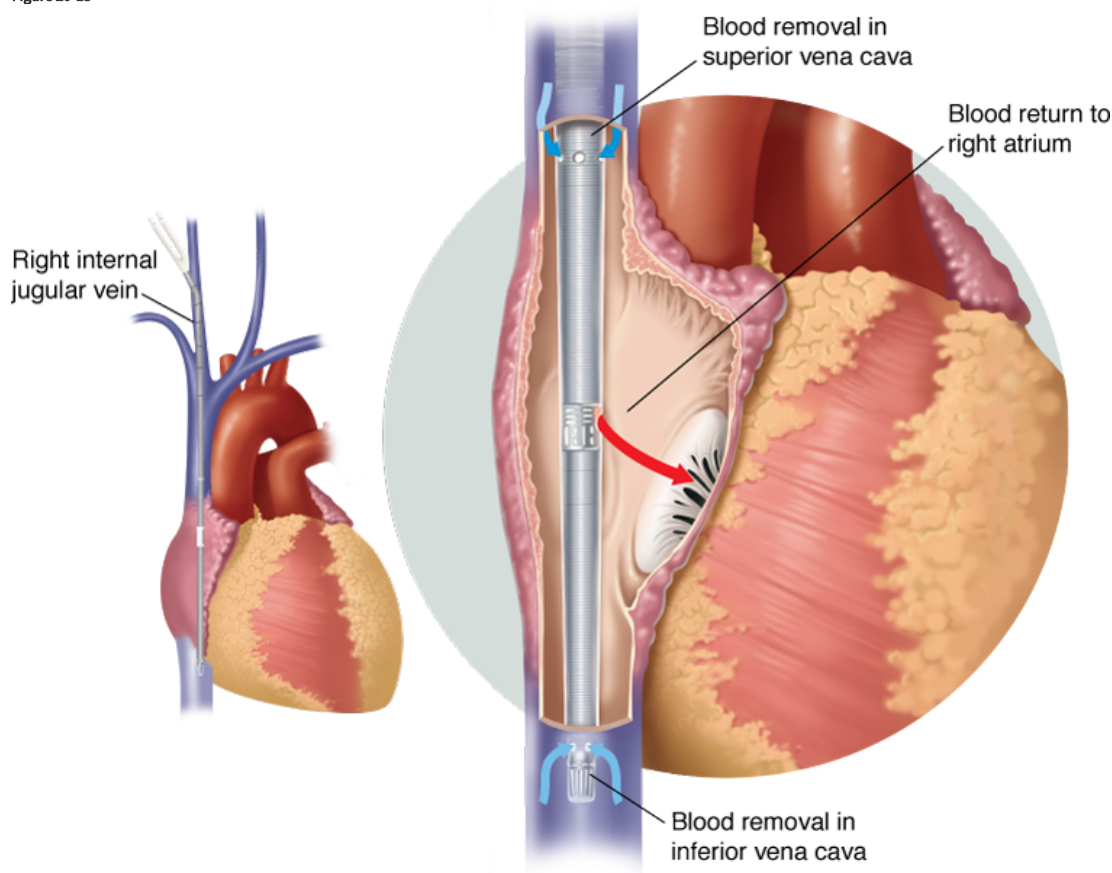
The use of ECLS in neonates has declined over the past 10 years as improved perinatal care and management techniques have advanced.^{82,83} Infantile respiratory distress syndrome has largely disappeared with the advent of surfactant and better lung-protective ventilation strategies, as well as changes in ventilator technology, such as the high-frequency oscillator. Meconium aspiration syndrome is also not as prevalent as in the past, and group B streptococcal infection has also declined. These diseases have been replaced with other infections and more complex infant illnesses. Congenital diaphragmatic hernia remains a large group for neonatal ECLS and survival in these patients remains only approximately 50% despite the many different management algorithms and approaches for surgical repair that have been introduced over time.^{84–89} The overall survival in neonatal ECLS patients is 69% over the past 10 years, with about 1000 infants receiving ECLS per year.⁹⁰ Based on ventilator setting and blood-gas data, there has been little change in severity over time based on oxygenation index (average: 45 to 50) or $P_{aO_2}/F_{I_{O_2}}$ ratio (average: 50–60 torr) for neonates, although a steady decline in pH and concomitant increase in P_{aCO_2} have been observed. A recent evaluation of the ELSO registry from 2000 to 2010 revealed a 2.5% decline in survival per year ($p < 0.01$).⁹¹

Although venoarterial access has been the mainstay in neonates, the fact that many infants with respiratory failure have good cardiac function makes them good candidates for venovenous support. Improved double-lumen, single-site cannulas that can be placed in the internal jugular vein have been developed, which allows for venovenous ECLS to be used in almost half of neonatal respiratory failure patients. Figure 20-13 shows a typical venovenous cannula that is currently widely used in larger children and adults. Figure 20-14 depicts a typical venovenous circuit. One problem that has been encountered with double-lumen cannulas has been difficulty with collapse of the drainage lumen if a high amount of venous “suction” is generated. This is potentially more of an issue with centrifugal pumps than with the older style semiocclusive, rollerhead devices.⁹² Development of wire-reinforced double-lumen, single cannulas has been spurred by the many centers that are changing to centrifugal equipment for its ease of use and ability to use shorter circuits with less priming volume. Although these cannulas have been very successful in larger patients, smaller sizes have encountered some difficulty with accurate placement, and clinical reports of unexplained tamponade occurring days after placement without well-identified myocardial perforations have surfaced. Newer models are expected in clinical trials soon. Current management is to try venovenous ECLS

Loading [Contrib]/a11y/accessibility-menu.js

support in almost any neonate without known complete cardiac collapse before implementing cardiac cannulation for venoarterial support. Successful use of venovenous support even in infants with primary respiratory failure and pre-ECLS need for vasoactive support has been well described.⁹³

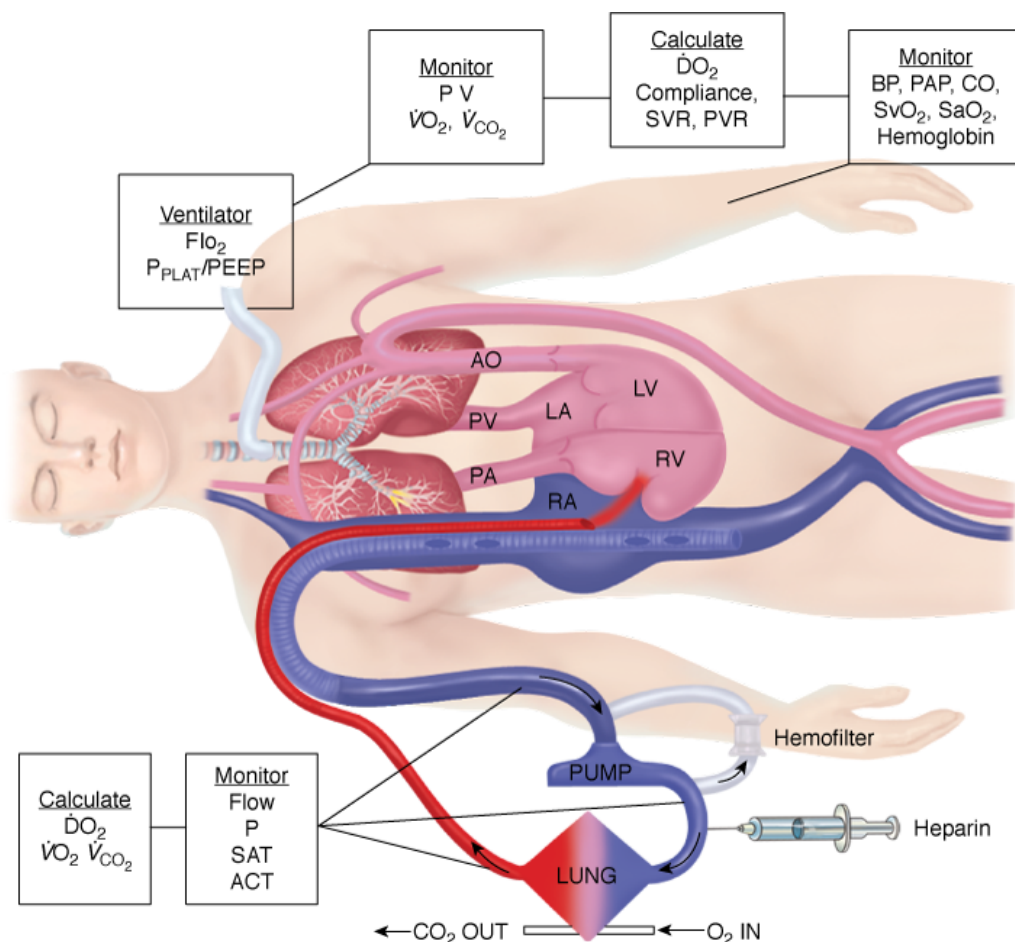
Figure 20-13



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com
Copyright © The McGraw-Hill Companies, Inc. All rights reserved.

Venovenous double-lumen cannula. To be properly positioned, the distal drainage port must sit in the inferior vena cava at the right-atrial junction and proximal drainage port in the superior vena cava/right-atrial junction. The inflow port will be directed at the tricuspid valve. (Photo courtesy of Avalon Laboratories, LLC, Rancho Dominguez CA.)

Figure 20-14



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com

Copyright © The McGraw-Hill Companies, Inc. All rights reserved.

Venovenous ECLS circuit. Boxes represent calculations and parameters that can be measured by monitoring site. *ACT*, activated clotting time; *BP*, arterial blood pressure; *CO*, cardiac output (estimated); *DO₂*, oxygen delivery; *P*, pressure; *PAP*, pulmonary artery pressure (only in patients with a pulmonary artery catheter in place); *PEEP*, positive end-expiratory pressure; *P_{PLAT}*, plateau pressure; *PVR*, pulmonary vascular resistance; *SaO₂*, arterial oxygen saturation; *Sat*, saturation of blood; *s $\bar{\text{v}}\text{O}_2$* , venous oxygen saturation; *SVR*, systemic vascular resistance; *V*, volume; $\dot{V}\text{CO}_2$, carbon dioxide; $\dot{V}\text{O}_2$, oxygen consumption. (Courtesy of Robert Bartlett, MD, with permission.)

Pediatric

As experience with ECLS has grown, the expansion to patient populations avoided in the past is nowhere as clearly seen as with pediatric patients. The “old days” when a healthy child would contract overwhelming pneumonia and be rescued with ECLS support are now a memory, as most pediatric ECLS patients today have underlying comorbidities in addition to their acute critical illness.^{94–101} The latest review of pediatric respiratory failure patients entered into the ELSO database was recently published.¹⁰² This report compared patients who received ECMO from 1993 to 2007. Survivors were noted to have a lower median body weight (9 vs. 9.9 kg), with a similar median age in survivors and nonsurvivors. Older children (ages 10 to 18 years) had lower survival (50%) compared to infants (57%), toddlers (61%), and children (55%). Although there was little change in survival over time, patients with comorbidities increased from 19% in 1993 to 47% in 2007. Renal failure, chronic lung disease, and congenital heart disease (two ventricles) formed the bulk of underlying comorbid conditions. Other conditions, which represent the changes in exclusion criteria over time, note that patients with cancer, solid-organ transplantation, immunodeficiencies, and even stem cell transplantation patients have received ECLS and survived. All these conditions would have been excluded as candidates for ECLS in the past. Nonetheless, children with no comorbidities have experienced improved survival with ECLS over time: 57% in 1993 to 72% in 2007. Another important factor in this summary of more than 3000 pediatric ECLS patients is that a significant decline in survival was not noted until patients reached a ventilator duration of longer than 14 days before ECMO. This is a large change from prior reports, which found that a ventilator duration of longer than 7 days was associated with worsening outcome. The use of high-frequency ventilation did not change over the period 1993 to 2007, and survival was not different between modes of ventilation. Patients who received high-frequency oscillation, however, did have a longer period of ventilation before ECMO than did those who received conventional mechanical ventilation (4.6 vs. 3 days). Nonsurvivors had a higher oxygenation index (48 vs. 42) and lower pH (7.27 vs. 7.31). Of note, pre-ECMO pH was progressively lower in both survivors and nonsurvivors as the years progressed. A trend in increased use of venovenous ECMO over years was also

Loading [Contrib]/a11y/accessibility-menu.js

observed, and venovenous ECMO was associated with improved outcome. As the ELSO registry gives little detail on severity of illness before ECMO, however, interpretations of outcome comparisons between venoarterial and venovenous ECMO can be difficult.

Table 20-2 shows the major categories of pediatric respiratory failure who have received ECLS. One difficulty in interpreting ELSO data is that many patients who are currently receiving ECLS have a combination of respiratory and cardiac failure, making assigning them to one of the three major groups in the ELSO registry (respiratory, cardiac, or emergency cardiopulmonary resuscitation) difficult. This skews the ability to accurately report results and is one major area where the ELSO registry is being renovated. Without data on severity of illness beyond the respiratory indices previously discussed, identifying risk factors for outcome or complications is difficult. A growing population where this is easily represented is patients in septic shock: these patients often have both respiratory and cardiac dysfunction, as well as injury to other organs.¹⁰³ Outcome of patients in septic shock is believed to be lower than in patients with single-organ failure, although good comparisons on a large scale have not been conducted. Better refinement of organ failure and severity of illness by the ELSO registry may allow more sophisticated analyses of these complex patients in the future. Debate (without resolution) continues as to the efficacy of ECLS in high-output failure versus low-cardiac output states, where it seems more intuitive that the additional oxygen delivery provided by ECLS may be of greater benefit to tissue oxygenation. In the Australian series of children with sepsis, central cannulation achieved significantly greater flow rates than did peripheral cannulation.⁶³ This observation requires confirmation in a larger series. As with neonates, lower pH and higher PaCO₂ values have been observed over time, with oxygenation index and PaO₂/FiO₂ remaining fairly constant (see Fig. 20-4). Despite the push for more venovenous support and better cannula availability, there does not seem to have been much of a change in the mode of ECLS support for respiratory failure patients. Although not statistically significant, an overview of data from the ELSO registry between 2000 and 2010 noted a slight increase in survival, 1.9% per year, despite the complexity of patients receiving ECLS.¹⁰⁴

Adult

An increase in the use of ECLS in adults is occurring with the advent of new technology, the CESAR trial, the H1N1 epidemic, and reports of successful use around the globe.^{15,105} Although most adult patients with respiratory support may be appropriate candidates for venovenous support, about one-third are managed with venoarterial cannulation. There has been an increase in venovenous support, however, over the past 2 years probably as a consequence of improved double-lumen cannulas and simpler, easier-to-use pumps and oxygenators.^{106,107} Over the past year, three-quarters of adult patients received venovenous ECLS via multiple sites or use of a double-lumen cannula. Chapter 21 discusses the use of venovenous support and the various devices that focus on gas exchange. Within the ELSO registry, overall survival in adult respiratory failure is similar to that in pediatric patients at 56%. One difficulty with both pediatric and adult patients is that the largest category reported to ELSO is “other,” which is obviously ill-defined and makes it difficult to identify exactly what type of disease these patients have. Similar to pediatric ECLS patients, adult survival between 2000 and 2010 has noted a slight (nonsignificant) increase in survival of 2.5% per year.

Cardiac

Although not the focus of this chapter, most patients requiring venoarterial support have cardiac failure, as a result of congenital heart defects, myocarditis, cardiomyopathy, cardiogenic shock, or before or after heart transplantation.^{77,106–120} Patients who receive ECLS emergently as a rescue mode of resuscitation from cardiac arrest (external cardiopulmonary resuscitation [ECPR]) are now an increasing population within the ELSO registry. Up to July 2011, 694 neonatal ECPR patients were reported, with 39% surviving to discharge; 1347 pediatric ECPR patients, with 40% survival; and 591 adult ECPR patients, with 29% surviving to discharge. The acceptance of advanced cardiac life support as a resuscitative tool is highlighted by recent statements from groups such as the American Heart Association: “To consider extracorporeal cardiopulmonary resuscitation [cardiopulmonary resuscitation] for in-hospital cardiac arrest refractory to initial resuscitation attempts if the condition leading to cardiac arrest is reversible or amenable to heart transplantation, if excellent conventional cardiopulmonary resuscitation has been performed after no more than several minutes of no-flow cardiac arrest, and if the institution is able to rapidly perform extracorporeal membrane oxygenation.”^{121–123}

Patient Management during Venoarterial Extracorporeal Life Support

Supporting patients who are receiving ECLS involves a mixture of providing adequate gas exchange, hemodynamic support, and minimizing risks of major complications such as bleeding and infection. As already discussed, gas exchange is controlled by a combination of blood flow to the oxygenator, surface area of the oxygenator, and sweep gas mixture. Oxygenation is predominantly influenced by the amount of blood flow to the oxygenator and carbon dioxide by the amount of sweep gas and surface area provided by the membrane in relation to the blood flow through the oxygenator. The measured oxygen saturation and carbon dioxide in the patient is a combination of the blood returning from the oxygenator and that being ejected by the patient’s left ventricle. The extent of hemodynamic support is provided by the amount of blood returning from the oxygenator to the patient, as well as the systemic vascular resistance, oxygen delivery, and extraction rates in the patient. Patients with poor cardiac function require more support from the ECLS circuit, and oxygen delivery can also be increased by provision of well-oxygenated blood returning from the ECLS as compared to that poorly oxygenated blood ejected from the left ventricle in patients with severe respiratory failure. Careful monitoring of the adequacy of ECLS support over time is important, because failure to reverse lactic or metabolic acidosis in the first 24 hours after ECLS is associated with poor outcome.^{124,125} As

centrifugal pump systems are becoming more widely used in ECLS, it is important to recognize that these systems are especially reliant on adequate preload in the patient for venous drainage and sensitive to afterload for arterial return. When a patient's systemic vascular resistance (SVR) increases, flow from the centrifugal circuit to the patient can be reduced unless the revolutions per minute are increased to maintain forward flow and overcome the added resistance being generated in the patient. Thus, an unexpected decrease in flow in the ECLS circuit may be the result of agitation, temperature decrease, or other factors that can increase SVR. Afterload reduction or other measures that reduce SVR may augment forward flow from the circuit. Likewise, reduction in SVR, as in sepsis, may indicate a need for increased flow to provide adequate oxygen delivery. There are some patients in whom tissue oxygenation is inadequate no matter what the flow provided; these patients are the most difficult to treat and have high mortality.

To maintain consistent patient support, most centers set daily goals for bedside personnel to follow with regards to the levels of oxygen, carbon dioxide, blood pressure, hemoglobin, and anticoagulation to be maintained.

Exposure to the ECLS circuit causes destruction of red blood cells and adherence of platelets, necessitating intermittent administration of blood products.¹²⁶ The underlying illness may also affect blood product production or increased consumption. It is conventional wisdom to maintain hemoglobin close to 15 g/dL to optimize oxygen content, although no studies have definitively shown that maintaining a high hematocrit enhances survival.^{127–129} The current debate regarding the risks and benefits of blood transfusion in critically ill patients holds true for ECLS: some clinicians limit blood exposure by maintaining hematocrit levels closer to 30%, whereas others maintain levels greater than 40%.^{130,131} The rheology of blood flowing through newly available cannulas, tubings, and oxygenators make this question a needed focus for future research to determine what is the “best” hematocrit during ECLS.

In a similar fashion, platelets levels have traditionally been maintained at more than 100,000, although there are numerous reports where patients have platelet levels of 20,000 or less without excessive bleeding. When bleeding does occur, attempts to increase platelet levels is a common practice, although, again, no definitive study shows that this step improves outcome.¹³² Other coagulation indices (prothrombin time, partial thromboplastin time, fibrinogen) are followed, and corrected with administration of replacement factors with plasma or cryoprecipitate.

As anticoagulation with heparin is standard practice to avoid clotting of the extracorporeal circuit, maintaining adequate heparinization while avoiding bleeding is a delicate balance. The activated clotting time, determined by a bedside test, is the standard measurement used for adjusting heparin dosage with ECLS.¹³³ Usually maintained at levels of 180 to 220 seconds, recent studies show that this test correlates poorly with measured heparin levels and indices of anticoagulation measured by other means. There is a continued need to develop more effective methods to assess how the level of anticoagulation relates to observed bleeding. The best method to reverse bleeding that is poorly controlled by simple administration of blood products or surgical intervention to establish hemostasis remains the most perplexing problem with ECLS today. Use of factor concentrates, such as VIIa and antithrombin, as well as monitoring of antithrombin, factor Xa, and thromboelastograph data are the current methods used by clinicians in an attempt to address bleeding and anticoagulation concerns.^{134–140} Bleeding remains one of the most common complications during ECLS. Lowering the heparin dose, even to the point of stopping heparin for 12 to 24 hours, has been successfully reported.¹⁴¹ Heparin-induced thrombocytopenia is another problem, leading to use of alternative agents such as argatroban or lepirudin.¹⁴² Bleeding has also been treated with agents that help stabilize clot, such as aminocaproic acid or aprotinin (which is no longer available in the United States).¹⁴³

Maintaining adequate fluid balance is another goal during ECLS. While diuretics remain the mainstay of fluid removal, many patients have renal insufficiency requiring hemofiltration or dialysis.^{144–148} The association between renal insufficiency and outcome has produced both reduced survival and unchanged outcomes in various reports.

Patients on ECLS are at risk for infection from changes in leukocyte function, nosocomial contamination of indwelling cannulas, and underlying critical illness.^{149–151} Although infection during ECLS has been reported, and may be associated with worsened survival, the early practice of “prophylactic” antibiotics has been fairly well abandoned. Periodic monitoring and treatment of documented infection is important, and successful recovery, even after nosocomial infection with organisms such as fungus, has been reported.

Satisfactory nutrition is important to promote healing during ECLS. Enteral nutrition is preferred, although parenteral support may be required if a patient cannot tolerate full enteral calories.^{152,153} The ability to tolerate feedings can be adversely affected by medications that reduce gut motility. Chief among these are narcotics.

One major difference in patient management between centers is the use of sedation.^{154,155} The growing movement to lessen sedation and keep patients more awake on ECLS produces benefit beyond the ability to monitor neurologic function. It also promotes the ability to adequately feed the patient, and maintain normal body movement to reduce peripheral edema and promote skin integrity. Minimizing sedation decreases the inherent risks of using increasing doses of sedative and analgesic agents secondary to toxicity or withdrawal. The new, more miniaturized circuits and improved

cannulas are one reason that more patients are being maintained wide-awake on ECLS, riding exercise bikes, or reading or eating ice cream (Fig. 20-15). Hopefully, this trend will continue.

Figure 20-15



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com
Copyright © The McGraw-Hill Companies, Inc. All rights reserved.

Ambulatory ECLS is becoming more common. (Photo courtesy of Ira Cheifetz MD, with permission.)

The optimal ventilator management during ECLS is debatable, although all agree that minimizing ventilator-induced lung injury is a goal.^{156–158} For the patient who is awake on venoarterial ECLS, because most gas exchange needs can be performed by the oxygenator, ventilator weaning to the point of extubation or minimal settings is a possibility. Some patients exhibit air-hunger or distress despite adequate oxygenation and ventilation provided by ECLS, and may require some pressure support or breathing assistance. Use of PEEP to maintain lung expansion may prevent atelectasis. Keeping inspiratory pressure of less than 30 cm H₂O is recommended. Promoting spontaneous breathing is important and recommended. Although use of high frequency ventilation is also described, this mode makes it difficult to keep patients awake and interpret actual tidal volumes. It is not commonly used unless attempts are being made to “push” the patient off ECLS secondary to complications.

Determining when a patient is ready for weaning is relatively simple during venoarterial ECLS. For patients with respiratory failure, improving oxygenation, clearing of the chest radiograph and improved tidal volumes at low pressure ventilation often herald recovery.^{156–158} For patients with hemodynamic compromise, the ability to maintain adequate blood pressure and tissue perfusion on lowered flows of ECLS may indicate time to wean.¹⁵⁹ Use of low-dose vasoactive agents may be needed to assist cardiac function and facilitate weaning. To truly assess the ability to wean off venoarterial support, most centers clamp the patient off support and observe gas exchange and hemodynamics with the patient separated from ECLS support. Echocardiography to evaluate cardiac performance may also be useful during this period.¹⁶⁰ If weaning is not tolerated, the cannulas can be unclamped and the patient continued on ECLS. During clamping, it is important to remember that a patient needs to remain heparinized and the cannulas “flushed” by releasing the clamps every few minutes to prevent clotting. Patience is required while awaiting organ recovery. A recent report noted no difference in survival for patients who required ECLS for less than 10 days compared to those maintained for more than 21 days. The impact of ventilation days before ECLS has also shown that survival can be obtained with durations thought “too long” in the past.^{161,162}

Outcome

Tables 20-1 and 20-2 list the overall survival for different patient groups. The greatest need is for long-term studies to assess neurodevelopmental outcome. Although short-term reports have found reasonable outcome, the impact of ECLS on school performance, risk for stroke later in life, and

quality of life is not well known.^{163,164} Older children and adults seem to have global neurologic function that matches that observed when they are discharged. The most in-depth reports of outcome have been in neonates with respiratory or cardiac failure.^{165–168} In the ELSO registry, neonates placed on ECLS for respiratory failure have an incidence of seizures of approximately 10%, although there is poor correlation with seizures noted during ECLS and overall neurologic outcome. Approximately 7% of neonates with respiratory failure are noted to have abnormal imaging studies (either hemorrhage or stroke). Two-thirds of neonatal respiratory failure survivors appear to have a normal neurodevelopmental outcome. Severe chronic respiratory disease in patients treated with ECMO is uncommon. Most authors report an incidence of bronchopulmonary dysplasia (defined as the need for oxygen beyond the first month of life) of 4% to 27%.¹⁶⁹ Most cases occurred in patients who had required extreme ventilator settings for more than 7 days before ECMO rescue. A follow-up report of neonates treated with ECMO and evaluated at 10 to 15 years post-ECMO found that although the ECMO patients had some impairment on pulmonary function testing, they had similar aerobic capacity and were able to reach similar anaerobic exercise goals as age-matched healthy controls.¹⁷⁰

Of 5000 pediatric respiratory ECMO patients listed in the Registry through July 2011, 9% had intracranial infarct or hemorrhage on computed tomography examination. Brain death occurred in 6% of the patients and another 6% had seizures. Long-term neurologic outcome data are sorely missing in the pediatric population. In one review of fifteen pediatric and four adult patients, 58% survived to discharge. Patients were evaluated by use of the pediatric cerebral performance category, which measures cognitive impairment, and the pediatric overall performance category, which measures functional morbidity. Overall, 64% of survivors had normal pediatric cerebral performance category scores, 27% had mild disabilities, and 9% had moderate cognitive disability. Functional morbidity was normal in 27%, while 45% had mild disability, 18% moderate disability, and 9% were severely disabled.¹⁷¹ In another small series of twenty-six patients followed 1 to 3 years after ECMO, 38% of pre-school-age children were described as normal and 31% exhibited abnormalities. Four patients (31%) who had prior neurologic dysfunction remained at baseline following ECMO. Among school-age children, 77% were described as normal by parental report. More specific neurologic follow-up in the pediatric age groups is needed.^{172,173}

Neurologic complications in cardiac patients who receive ECMO parallel that of respiratory failure patients: 49% developed brain death, 3% had intracranial infarct, and 6% had intracranial hemorrhage. Because many cardiac patients are in a state of prolonged low cardiac output or sudden cardiac arrest before ECMO, the ability to assess neurologic function once ECMO is instituted is vitally important. Paralysis and sedation should be minimized until neurologic examination can be performed. This information is especially important in patients who are being listed for transplantation, so as to avoid transplanting a viable organ into an inappropriate recipient. Among neonatal patients placed on ECLS for cardiac dysfunction, most have congenital heart disease. Outcome of survivors reveals that approximately 50% have normal neurodevelopmental outcome. New reports of neurodevelopmental outcome in patients with congenital heart disease reveal abnormalities, which were previously unsuspected. How many of the post-ECLS abnormalities are the result of the underlying condition, pre-ECLS events, or the result of ECLS itself, cannot be ascertained.¹⁷² In a recent 2-year follow-up study of ECLS performed in patients younger than 5 years of age, 46% survived to discharge and 41% were alive at the 2-year assessment.¹⁷⁴ Neurodevelopmental concerns were identified in most survivors, with a mean mental score of 73 ± 16 , mental delay in 50% of survivors, and motor or sensory disability in 12%.

In an evaluation of patients who had undergone ECLS for resuscitation during cardiopulmonary resuscitation, 22% of patients had acute neurologic injury, defined as brain death, brain infarction, or intracranial hemorrhage identified by ultrasound or computerized tomography imaging. Brain death occurred in 11%, cerebral infarction in 7%, and intracranial hemorrhage in 7%. The in-hospital mortality rate in patients with acute neurologic injury was 89%. During ECMO, neurologic injury was associated with ECMO complications such as pulmonary hemorrhage, dialysis use, and cardiopulmonary resuscitation. Pre-ECMO factors, including cardiac disease and pH greater than 6.8, were associated with decreased odds of neurologic injuries. In a review of adults resuscitated from cardiac arrest with ECLS, overall survival was 27%, with brain death occurring in 28% of nonsurvivors. Pre-ECLS factors, such as higher PaO₂, lower PaCO₂, ECLS finding of pH less than 7.2, need for renal replacement therapy, and development of hyperbilirubinemia or central nervous system injury on ECLS, were associated with poor outcome.^{175,176}

Among adult patients who received venovenous ECLS for respiratory compromise, a follow-up study of twenty-one patients noted that pulmonary function was at least 80% of normal. Quality-of-life measurements revealed that patients were adversely affected as compared to normal, although most had returned to work. Several other studies also note good recovery after ECLS survival.¹⁷⁷ Follow-up studies of patients from the CESAR trial will add much in terms of neurodevelopmental function in adult ECLS survivors.

Longer-term evaluation of patients surviving ECLS and comparison to patients with similar disease severity and diagnosis is imperative to adequately interpret neurologic outcome.^{178–183}

The Future

The current extension of ECLS systems to older pediatric and adult patients in various clinical settings highlights the changes that have occurred in this field. With the development of new cardiac support devices applicable to pediatrics, may further expand the use of ECLS techniques. Additionally, the development of small, portable systems for cardiopulmonary resuscitation

may herald a new age of ECLS and make interhospital transport easier, safer, and more available.^{184–186} Single-site, double-lumen catheters for venovenous ECMO may obviate the risks of arterial cannulation and offer the benefit of requiring only one surgical site for venous access. Heparin-bonded circuits may decrease the need for systemic anticoagulation and the risk of hemorrhagic complications.^{92,184,187} Given the myriad adverse events associated with mechanical ventilation, the thought that extracorporeal gas exchange may obviate the need for intubation in respiratory failure has been suggested. Until the day when medical science may make the need for ECLS obsolete, research into ways to make it safer and more efficient should continue.¹⁸⁸

References

1. Clowes GH Jr, Hopkins AL, Neville WE. An artificial lung dependent upon diffusion of [oxygen](#) and carbon dioxide through plastic membranes. *J Thorac Surg*. 1956;32:630–637. [[PubMed: 13368241](#)]

2. Gille JP. [Respiratory support by extracorporeal circulation with a membrane artificial lung] (author's transl). *Bull Physiopathol Respir (Nancy)*. 1974;10:373–410. [[PubMed: 4605920](#)]

3. Hill JD, Gibbon JH Jr. Part I. The development of the first successful heart-lung machine. *Ann Thorac Surg*. 1982;34:337–341. [[PubMed: 7052001](#)]

4. Page US, Bigelow JC, Carter CR, Swank RL. Emboli (debris) produced by bubble oxygenators. Removal by filtration. *Ann Thorac Surg*. 1974;18:164–171. [[PubMed: 4621031](#)]

5. Bartlett RH, Gazzaniga AB, Jefferies MR, et al. Extracorporeal membrane oxygenation (ECMO) cardiopulmonary support in infancy. *Trans Am Soc Artif Intern Organs*. 1976;22:80–93. [[PubMed: 951895](#)]

6. Kolobow T, Bowman RL. Construction and evaluation of an alveolar membrane artificial heart-lung. *Trans Am Soc Artif Intern Organs*. 1963;9:238–243. [[PubMed: 14034415](#)]

7. Heiss KF, Bartlett RH. Extracorporeal membrane oxygenation: an experimental protocol becomes a clinical service. *Adv Pediatr*. 1989;36:117–135. [[PubMed: 2675567](#)]

8. Short BL, Pearson GD. Neonatal extracorporeal membrane oxygenation: a review. *J Intensive Care Med*. 1986;1:48.

9. Stork E. Extracorporeal membrane oxygenation in the newborn and beyond. *Clin Perinatol*. 1988;15:815–829. [[PubMed: 3061700](#)]

10. Stolar CJ, Dillon PW. Extracorporeal membrane oxygenation for neonatal respiratory failure. *Surg Annu*. 1987;19:111–122. [[PubMed: 3547704](#)]

11. O'Rourke PP, Crone RK, Vacanti JP, et al. Extracorporeal membrane oxygenation and conventional medical therapy in neonates with persistent pulmonary hypertension of the newborn: a prospective randomized study. *Pediatrics*. 1989;84:957–963. [[PubMed: 2685740](#)]

12. Bluhm R. The epistemology and ethics of chronic disease research: further lessons from ECMO. *Theor Med Bioeth*. 2010;31:107–122. [[PubMed: 20333473](#)]

13. UK collaborative randomised trial of neonatal extracorporeal membrane oxygenation. UK Collaborative ECMO Trail Group. *Lancet*. 1996;348:75–82.

14. Dalton HJ. Extracorporeal life support in the new millennium: forging ahead or fading out? *New Horizons*. 1999;7:414–432.

15. Kolobow T. An update on adult extracorporeal membrane oxygenation-extracorporeal CO₂ removal. *ASAIO Trans*. 1988;34:1004–1005. [[PubMed: 3146338](#)]

16. Snider MT, Campbell DB, Kofke WA, et al. Venovenous perfusion of adults and children with severe acute respiratory distress syndrome. The Pennsylvania State University experience from 1982–1987. *ASAIO Trans*. 1988;34:1014–1020. [[PubMed: 3219247](#)]

17. Hill JD, O'Brien TG, Murray JJ, et al. Prolonged extracorporeal oxygenation for acute post-traumatic respiratory failure (shock-lung syndrome). Use of the Bramson membrane lung. *N Engl J Med*. 1972;286:629–634. [[PubMed: 5060491](#)]

18. Zapol WM, Snider MT, Hill JD, et al. Extracorporeal membrane oxygenation in severe acute respiratory failure. A randomized prospective study. *AMA*. 1978;242:3103–3106. [[PubMed: 490805](#)]

19. Kolla S, Awad SS, Rich PB, et al. Extracorporeal life support for 100 adult patients with severe respiratory failure. *Ann Surg.* 1997;226:544–564; discussion 565–566. [\[PubMed: 9351722\]](#)

20. Gattinoni L, Kolobow T, Tomlinson T, et al. Low-frequency positive pressure ventilation with extracorporeal carbon dioxide removal (LFPPV-ECCO2R): an experimental study. *Anesth Analg.* 1978;57:470–477. [\[PubMed: 30341\]](#)

21. Dalton HJ, Heulitt MJ. Extracorporeal membrane oxygenation. Invited Review. *Respir Care.* 1998;43:966–977.

22. Hemmila MR, Rowe SA, Boules TN, et al. Extracorporeal life support for severe acute respiratory distress syndrome in adults. *Ann Surg.* 2004;240:595–605; discussion 605–607. [\[PubMed: 15383787\]](#)

23. International Registry of the Extracorporeal Life Support Organization, Ann Arbor, Michigan; July 2011.

24. O'Rourke PP, Stolar CJ, Zwischenberger JB, et al. Extracorporeal membrane oxygenation: support for overwhelming pulmonary failure in the pediatric population. Collective experience from the extracorporeal life support organization. *J Pediatr Surg.* 1993;28:523–528; discussion 528–529. [\[PubMed: 8483064\]](#)

25. Timmons OD, Dean JM, Vernon DD. Mortality rates and prognostic variables in children with adult respiratory distress syndrome. *J Pediatr.* 1991;119:896–899. [\[PubMed: 1960603\]](#)

26. Timmons OD, Havens PL, Fackler JC. Predicting death in pediatric patients with acute respiratory failure. Pediatric Critical Care Study Group. Extracorporeal Life Support Organization. *Chest.* 1995;108:789–797. [\[PubMed: 7656635\]](#)

27. Green TP, Timmons OD, Fackler JC, et al. The impact of extracorporeal membrane oxygenation on survival in pediatric patients with acute respiratory failure. Pediatric Critical Care Study Group. *Crit Care Med.* 1996;24:323–329. [\[PubMed: 8605808\]](#)

28. Arnold JH, Anas NG, Luckett P, et al. High-frequency oscillatory ventilation in pediatric respiratory failure: a multicenter experience. *Crit Care Med.* 2000;28:3913–3919. [\[PubMed: 11153635\]](#)

29. Dobyns EL, Anas NG, Fortenberry JD, et al. Interactive effects of high-frequency oscillatory ventilation and inhaled nitric oxide in acute hypoxemic respiratory failure in pediatrics. *Crit Care Med.* 2002;30:2425–2429. [\[PubMed: 12441749\]](#)

30. Ventre KM, Arnold JH. High frequency oscillatory ventilation in acute respiratory failure. *Paediatr Respir Rev.* 2004;5:323–332. [\[PubMed: 15531258\]](#)

31. Kavanagh BP. Prone positioning in children with ARDS: positive reflections on a negative clinical trial. *JAMA.* 2005;294:248–250. [\[PubMed: 16014601\]](#)

32. Peek GJ, Mugford M, Tiruvoipati R, et al. Efficacy and economic assessment of conventional ventilatory support versus extracorporeal membrane oxygenation for severe adult respiratory failure (CESAR): a multicentre randomised controlled trial. *Lancet.* 2009;374:1351–1363. [\[PubMed: 19762075\]](#)

33. Peek GJ, Elbourne D, Mugford M, et al. Randomised controlled trial and parallel economic evaluation of conventional ventilatory support versus extracorporeal membrane oxygenation for severe adult respiratory failure (CESAR). *Health Technol Assess.* 2010;14:1–46. [\[PubMed: 20642916\]](#)

34. Morris AH, Hirshberg E, Miller RR 3rd, et al. Counterpoint: efficacy of extracorporeal membrane oxygenation in 2009 influenza A(H1N1): sufficient evidence? *Chest.* 2010;138:778–781; discussion 782–784. [\[PubMed: 20923797\]](#)

35. Park PK, Dalton HJ, Bartlett RH. Point: efficacy of extracorporeal membrane oxygenation in 2009 influenza A(H1N1): sufficient evidence? *Chest.* 2010;138:776–778. [\[PubMed: 20923796\]](#)

36. Zwischenberger JB, Lynch JE. Will CESAR answer the adult ECMO debate? *Lancet.* 2009;374:1307–1308. [\[PubMed: 19762074\]](#)

37. Davies A, Jones D, Bailey M, et al. Extracorporeal membrane oxygenation for 2009 influenza A(H1N1) acute respiratory distress syndrome. *JAMA.* 2009;302:1888–1895. [\[PubMed: 19822628\]](#)

38. Dominguez-Cherit G, Lapinsky SE, Macias AE, et al. Critically ill patients with 2009 influenza A(H1N1) in Mexico. *JAMA.* 2009;302:1880–1887. [\[PubMed: 19822626\]](#)

39. Kumar A, Zarychanski R, Pinto R, et al. Critically ill patients with 2009 influenza A(H1N1) infection in Canada. *JAMA*. 2009;302:1872–1879. [PubMed: 19822627]

40. ELSO: www.ELSO.med.umich.edu.

41. Beck R, Anderson KD, Pearson GD, et al. Criteria for extracorporeal membrane oxygenation in a population of infants with persistent pulmonary hypertension of the newborn. *J Pediatr Surg*. 1986;21:297–302. [PubMed: 3084751]

42. Marsh TD, Wilkerson SA, Cook LN. Extracorporeal membrane oxygenation selection criteria: partial pressure of arterial oxygen versus alveolar-arterial oxygen gradient. *Pediatrics*. 1988;82:162–166. [PubMed: 3399289]

43. Nading JH. Historical controls for extracorporeal membrane oxygenation in neonates. *Crit Care Med*. 1989;17:423–425. [PubMed: 2707012]

44. Ortiz RM, Cilley RE, Bartlett RH. Extracorporeal membrane oxygenation in pediatric respiratory failure. *Pediatr Clin North Am*. 1987;34:39–46. [PubMed: 3808772]

45. Peters MJ, Tasker RC, Kiff KM, et al. Acute hypoxemic respiratory failure in children: case mix and the utility of respiratory severity indices. *Intensive Care Med*. 1998;24:699–705. [PubMed: 9722040]

46. Trachsel D, McCrindle BW, Nakagawa S, Bohn D. Oxygenation index predicts outcome in children with acute hypoxemic respiratory failure. *Am J Respir Crit Care Med*. 2005;172:206–211. [PubMed: 15817802]

47. Carcillo JA, Fields AI. Clinical practice parameters for hemodynamic support of pediatric and neonatal patients in septic shock. *Crit Care Med*. 2002;30:1365–1378. [PubMed: 12072696]

48. Gaies MG, Gurney JG, Yen AH, et al. Vasoactive-inotropic score as a predictor of morbidity and mortality in infants after cardiopulmonary bypass. *Pediatr Crit Care Med*. 2010;11:234–238. [PubMed: 19794327]

49. Kulik TJ, Moler FW, Palmisano JM, et al. Outcome-associated factors in pediatric patients treated with extracorporeal membrane oxygenator after cardiac surgery. *Circulation*. 1996;94:II63–II68.

50. MacLaren G, Pellegrino V, Butt W, et al. Successful use of ECMO in adults with life-threatening infections. *Anaesth Intensive Care*. 2004;32:707–710. [PubMed: 15535500]

51. Andrews AF, Klein MD, Toomasian JM, et al. Venovenous extracorporeal membrane oxygenation in neonates with respiratory failure. *J Pediatr Surg*. 1983;18:339–346. [PubMed: 6620071]

52. Pettignano R, Fortenberry JD, Heard ML, et al. Primary use of the venovenous approach for extracorporeal membrane oxygenation in pediatric acute respiratory failure. *Pediatr Crit Care Med*. 2003;4:291–298. [PubMed: 12831409]

53. Van Meurs KP, Mikesell GT, Seale WR, et al. Maximum blood flow rates for arterial cannulae used in neonatal ECMO. *ASAIO Trans*. 1990;36:M679–M681.

54. Karl TR, Iyer KS, Sano S, Mee RB. Infant ECMO cannulation technique allowing preservation of carotid and jugular vessels. *Ann Thorac Surg*. 1990;50:488–489. [PubMed: 2400278]

55. Adolph V, Bonis S, Falterman K, Arensman R. Carotid artery repair after pediatric extracorporeal membrane oxygenation. *J Pediatr Surg*. 1990;25:867–869; discussion 869–870. [PubMed: 2205714]

56. Crombleholme TM, Adzick NS, deLorimier AA, et al. Carotid artery reconstruction following extracorporeal membrane oxygenation. *Am J Dis Child*. 1990;144:872–874. [PubMed: 2198805]

57. Desai SA, Stanley C, Gringlas M, et al. Five-year follow-up of neonates with reconstructed right common carotid arteries after extracorporeal membrane oxygenation. *J Pediatr*. 1999;134:428–433. [PubMed: 10190916]

58. Schumacher RE, Barks JD, Johnston MV, et al. Right-sided brain lesions in infants following extracorporeal membrane oxygenation. *Pediatrics*. 1988;82:155–161. [PubMed: 3399288]

Loading [Contrib]/a11y/accessibility-menu.js

59. Pettignano R, Labuz M, Gauthier TW, et al. The use of cephalad cannulae to monitor jugular venous [oxygen](#) content during extracorporeal membrane oxygenation. *Crit Care*. 1997;1:95–99. [[PubMed: 11056701](#)]
60. Skarsgard ED, Salt DR, Lee SK. Venovenous extracorporeal membrane oxygenation in neonatal respiratory failure: does routine, cephalad jugular drainage improve outcome? *J Pediatr Surg*. 2004;39:672–676. [[PubMed: 15136997](#)]
61. Engel M, Dalton HJ. Expanding Horizons in extracorporeal life support. Invited review. *J Resp. Dis for Pediatricians*. 2003;5:115–125.
62. Pennington DG, Merjavy JP, Codd JE, et al. Extracorporeal membrane oxygenation for patients with cardiogenic shock. *Circulation*. 1984;70:1130–1137.
63. Maclaren G, Butt W, Best D, et al. Extracorporeal membrane oxygenation for refractory septic shock in children: one institution's experience. *Pediatr Crit Care Med*. 2007;8:447–451. [[PubMed: 17693912](#)]
64. MacLaren G, Butt W, Best D, Donath S. Central extracorporeal membrane oxygenation for refractory pediatric septic shock. *Pediatr Crit Care Med*. 2011;12:133–136. [[PubMed: 20453704](#)]
65. Chen YS, Ko WJ, Lin FY. Insertion of percutaneous ECMO cannula. *Am J Emerg Med*. 2000;18:184–185. [[PubMed: 10750927](#)]
66. Foley DS, Swaniker F, Pranikoff T, et al. Percutaneous cannulation for pediatric venovenous extracorporeal life support. *J Pediatr Surg*. 2000;35:943–947. [[PubMed: 10873041](#)]
67. Rich PB, Awad SS, Crotti S, et al. A prospective comparison of atrio-femoral and femoro-atrial flow in adult venovenous extracorporeal life support. *J Thorac Cardiovasc Surg*. 1998;116:628–632. [[PubMed: 9766592](#)]
68. Greason KL, Hemp JR, Maxwell JM, et al. Prevention of distal limb ischemia during cardiopulmonary support via femoral cannulation. *Ann Thorac Surg*. 1995;60:209–210. [[PubMed: 7598601](#)]
69. Moazami N, Moon MR, Lawton JS, et al. Axillary artery cannulation for extracorporeal membrane oxygenator support in adults: an approach to minimize complications. *J Thorac Cardiovasc Surg*. 2003;126:2097–2098. [[PubMed: 14688737](#)]
70. SciMed Membrane Oxygenators. Instruction manual. Minneapolis, MN: SciMed Life Systems; 2000.
71. Kawahito S, Motomura T, Glueck J, Nose Y. Development of a new hollow fiber silicone membrane oxygenator for ECMO: the recent progress. *Ann Thorac Cardiovasc Surg*. 2002;8:268–274. [[PubMed: 12472408](#)]
72. Habashi NM, Borg UR, Reynolds HN. Low blood flow extracorporeal carbon dioxide removal (ECCO2R): a review of the concept and a case report. *Intensive Care Med*. 1995;21:594–597. [[PubMed: 7593903](#)]
73. Lawson DS, Lawson AF, Walczak R, et al. North American neonatal extracorporeal membrane oxygenation (ECMO) devices and team roles: 2008 survey results of Extracorporeal Life Support Organization (ELSO) centers. *J Extra Corpor Technol*. 2008;40:166–174. [[PubMed: 18853828](#)]
74. Liebold A, Reng CM, Philipp A, et al. Pumpless extracorporeal lung assist—experience with the first 20 cases. *Eur J Cardiothorac Surg*. 2000;17:608–613. [[PubMed: 10814928](#)]
75. Seo T, Ando H, Ito T, et al. Development of disposable self-regulating blood pumps and automatically-controlled portable extracorporeal membrane oxygenation systems for neonatal extracorporeal membrane oxygenation. *Artif Organs*. 2003;27:192–198. [[PubMed: 12580779](#)]
76. Secker-Walker JS, Edmonds JF, Spratt EH, Conn AW. The source of coronary perfusion during partial bypass for extracorporeal membrane oxygenation (ECMO). *Ann Thorac Surg*. 1976;21:138–143. [[PubMed: 1267911](#)]
77. Rogers AJ, Trento A, Siewers RD, et al. Extracorporeal membrane oxygenation for postcardiotomy cardiogenic shock in children. *Ann Thorac Surg*. 1989;47:903–906. [[PubMed: 2757447](#)]
78. Seib PM, Faulkner SC, Erickson CC, et al. Blade and balloon atrial septostomy for left heart decompression in patients with severe ventricular dysfunction on extracorporeal membrane oxygenation. *Catheter Cardiovasc Interv*. 1999;46:179–186. [[PubMed: 10348539](#)]

79. Kolobow T, Spragg RG, Pierce JE. Massive pulmonary infarction during total cardiopulmonary bypass in unanesthetized spontaneously breathing lambs. *Int J Artif Organs*. 1981;4:76–81. [\[PubMed: 6792084\]](#)
80. Mims BC. Physiologic rationale of Sv-O₂ monitoring. *Crit Care Nurs Clin North Am*. 1989;1:619–628. [\[PubMed: 2684246\]](#)
81. Thomas TH, Price R, Ramaciotti C, et al. Echocardiography, not chest radiography, for evaluation of cannula placement during pediatric extracorporeal membrane oxygenation. *Pediatr Crit Care Med*. 2009;10:56–59. [\[PubMed: 19057430\]](#)
82. Keckler SJ, Laituri CA, Ostlie DJ, St Peter SD. A review of venovenous and venoarterial extracorporeal membrane oxygenation in neonates and children. *Eur J Pediatr Surg*. 2010;20:1–4. [\[PubMed: 19746333\]](#)
83. Tulenko DR. An update on ECMO. *Neonatal Netw*. 2004;23:11–18. [\[PubMed: 15317374\]](#)
84. Connors RH, Tracy T Jr, Bailey PV, et al. Congenital diaphragmatic hernia repair on ECMO. *J Pediatr Surg*. 1990;25:1043–1046; discussion 1046–1047. [\[PubMed: 2262855\]](#)
85. Doyle NM, Lally KP. The CDH Study Group and advances in the clinical care of the patient with congenital diaphragmatic hernia. *Semin Perinatol*. 2004;28:174–184. [\[PubMed: 15283097\]](#)
86. Geggel RL, Murphy JD, Langleben D, et al. Congenital diaphragmatic hernia: arterial structural changes and persistent pulmonary hypertension after surgical repair. *J Pediatr*. 1985;107:457–464. [\[PubMed: 4032138\]](#)
87. Levin DL. Congenital diaphragmatic hernia: a persistent problem. *J Pediatr*. 1987;111:390–392. [\[PubMed: 3625406\]](#)
88. Tibboel D, Bos AP, Pattenier JW, et al. Pre-operative stabilisation with delayed repair in congenital diaphragmatic hernia. *Z Kinderchir*. 1989;44:139–143. [\[PubMed: 2750338\]](#)
89. Van Meurs KP, Newman KD, Anderson KD, Short BL. Effect of extracorporeal membrane oxygenation on survival of infants with congenital diaphragmatic hernia. *J Pediatr*. 1990;117:954–960.
90. ECMO Registry of the Extracorporeal Life Support Organization, Ann Arbor, MI: July 2011.
91. Dalton HJ, Garcia-Filion P. Unpublished data. 2010(a).
92. Yamagishi T, Kunimoto F, Isa Y, et al. Clinical results of extracorporeal membrane oxygenation (ECMO) support for acute respiratory failure: a comparison of a centrifugal pump ECMO with a roller pump ECMO. *Surg Today*. 2004;34:209–213. [\[PubMed: 14999531\]](#)
93. Roberts N, Westrope C, Pooboni SK, et al. Venovenous extracorporeal membrane oxygenation for respiratory failure in inotrope dependent neonates. *ASAIO J*. 2003;49:568–571. [\[PubMed: 14524566\]](#)
94. Fortenberry JD, Meier AH, Pettignano R, et al. Extracorporeal life support for posttraumatic acute respiratory distress syndrome at a children's medical center. *J Pediatr Surg*. 2003;38:1221–1226. [\[PubMed: 12891497\]](#)
95. Gow KW, Heiss KF, Wulkan ML, et al. Extracorporeal life support for support of children with malignancy and respiratory or cardiac failure: the extracorporeal life support experience. *Crit Care Med*. 2009;37:1308–1316. [\[PubMed: 19242331\]](#)
96. Hebbbar KB, Petrillo-Albarano T, Coto-Puckett W, et al. Experience with use of extracorporeal life support for severe refractory status asthmaticus in children. *Crit Care*. 2009;13:R29.
97. Linden V, Karlen J, Olsson M, et al. Successful extracorporeal membrane oxygenation in four children with malignant disease and severe *Pneumocystis carinii* pneumonia. *Med Pediatr Oncol*. 1999;32:25–31. [\[PubMed: 9917749\]](#)
98. O'Rourke PP, Crone RK. Pediatric applications of extracorporeal membrane oxygenation. *J Pediatr*. 1990;116:393–394. [\[PubMed: 2308029\]](#)
99. Sheridan RL, Schnitzer JJ. Management of the high-risk pediatric burn patient. *J Pediatr Surg*. 2001;36:1308–1312. [\[PubMed: 11479883\]](#)
100. Szocik J, Rudich S, Csete M. ECMO resuscitation after massive pulmonary embolism during liver transplantation. *Anesthesiology*. 2002;97:763–767. [\[PubMed: 12219507\]](#)

101. Thiagarajan RR, Roth SJ, Margossian S, et al. Extracorporeal membrane oxygenation as a bridge to cardiac transplantation in a patient with cardiomyopathy and hemophilia A. *Intensive Care Med.* 2003;29:985–988. [PubMed: 12719805]
102. Zabrocki LA, Brogan TV, Statler KD, et al. Extracorporeal membrane oxygenation for pediatric respiratory failure: survival and predictors of mortality. *Crit Care Med.* 2011;39:364–370. [PubMed: 20959787]
103. Beca J, Butt W. Extracorporeal membrane oxygenation for refractory septic shock in children. *Pediatrics.* 1994;93:726–729. [PubMed: 8165069]
104. Dalton HJ, Garcia-Filion P. Unpublished data. 2011.
105. Cordell-Smith JA, Roberts N, Peek GJ, Firmin RK. Traumatic lung injury treated by extracorporeal membrane oxygenation (ECMO). *Injury.* 2006;37:29–32. [PubMed: 16243331]
106. Alpard SK, Zwischenberger JB, Tao W, et al. Reduced ventilator pressure and improved P/F ratio during percutaneous arteriovenous carbon dioxide removal for severe respiratory failure. *Ann Surg.* 1999;230:215–224. [PubMed: 10450736]
107. Sivarajan VB, Best D, Brizard CP, et al. Improved outcomes of paediatric extracorporeal support associated with technology change. *Interact Cardiovasc Thorac Surg.* 2010;11:400–405. [PubMed: 20634272]
108. International Registry of the Extracorporeal Life Support Organization. Ann Arbor, University of Michigan: July 2009.
109. Chaturvedi RR, Macrae D, Brown KL, et al. Cardiac ECMO for biventricular hearts after paediatric open heart surgery. *Heart.* 2004;90:545–551. [PubMed: 15084554]
110. Dalton HJ, Siewers RD, Fuhrman BP, et al. Extracorporeal membrane oxygenation for cardiac rescue in children with severe myocardial dysfunction. *Crit Care Med.* 1993;21:1020–1028. [PubMed: 8319459]
111. del Nido PJ, Armitage JM, Fricker FJ, et al. Extracorporeal membrane oxygenation support as a bridge to pediatric heart transplantation. *Circulation.* 1994;90:II66–II69.
112. Duncan BW, Hraska V, Jonas RA, et al. Mechanical circulatory support in children with cardiac disease. *J Thorac Cardiovasc Surg.* 1999;117:529–542. [PubMed: 10047657]
113. Galantowica M, Stolar CJ. Extracorporeal membrane oxygenation for peri-operative support in pediatric heart transplantation. Extracorporeal Life Support Organization Charter Meeting. Ann Arbor, MI: 1989:48A.
114. Kanter KR, Pennington G, Weber TR, et al. Extracorporeal membrane oxygenation for postoperative cardiac support in children. *J Thorac Cardiovasc Surg.* 1987;93:27–35. [PubMed: 3796029]
115. Klein MD, Shaheen KW, Whittlesey GC, et al. Extracorporeal membrane oxygenation for the circulatory support of children after repair of congenital heart disease. *J Thorac Cardiovasc Surg.* 1990;100:498–505. [PubMed: 2214828]
116. Kolovos NS, Bratton SL, Moler FW, et al. Outcome of pediatric patients treated with extracorporeal life support after cardiac surgery. *Ann Thorac Surg.* 2003;76:1435–1441; discussion 1441–1442. [PubMed: 14602263]
117. Rajagopal SK, Almond CS, Laussen PC, et al. Extracorporeal membrane oxygenation for the support of infants, children, and young adults with acute myocarditis: a review of the Extracorporeal Life Support Organization registry. *Crit Care Med.* 2010;38:382–387. [PubMed: 19789437]
118. Raymond TT, Cunyngnam CB, Thompson MT, et al. Outcomes among neonates, infants, and children after extracorporeal cardiopulmonary resuscitation for refractory in-hospital pediatric cardiac arrest: a report from the National Registry of Cardiopulmonary Resuscitation. *Pediatr Crit Care Med.* 2010;11:362–371. [PubMed: 19924027]
119. Undar A, McKenzie ED, McGarry MC, et al. Outcomes of congenital heart surgery patients after extracorporeal life support at Texas Children's Hospital. *Artif Organs.* 2004;28:963–966. [PubMed: 15385006]
120. Weinhaus L, Canter C, Noetzel M, et al. Extracorporeal membrane oxygenation for circulatory support after repair of congenital heart defects. *Ann Thorac Surg.* 1989;48:206–212. [PubMed: 2669646]

121. del Nido PJ, Dalton HJ, Thompson AE, Siewers RD. Extracorporeal membrane oxygenator rescue in children during cardiac arrest after cardiac surgery. *Circulation*. 1992;86:II300–II304.
122. Kleinman ME, Chameides L, Schexnayder SM, et al. Part 14: pediatric advanced life support: 2010 American Heart Association guidelines for cardiopulmonary resuscitation and emergency cardiovascular care. *Circulation*. 2010;122:S876–S908.
123. Morris MC, Wernovsky G, Nadkarni VM. Survival outcomes after extracorporeal cardiopulmonary resuscitation instituted during active chest compressions following refractory in-hospital pediatric cardiac arrest. *Pediatr Crit Care Med*. 2004;5:440–446. [PubMed: 15329159]
124. Kumar TK, Zurawski D, Dalton H, et al. Extracorporeal membrane oxygenation in postcardiotomy patients: factors influencing outcome. *J Thorac Cardiovasc Surg*. 2010;140:330–336.e2. [PubMed: 20637917]
125. Tajik M, Cardarelli MG. Extracorporeal membrane oxygenation after cardiac arrest in children: what do we know? *Eur J Cardiothorac Surg*. 2008;33:409–417. [PubMed: 18206379]
126. Zavadil DP, Stammers AH, Willett LD, et al. Hematological abnormalities in neonatal patients treated with extracorporeal membrane oxygenation (ECMO). *J Extra Corpor Technol*. 1998;30:83–90. [PubMed: 10182118]
127. Gilbert EM, Haupt MT, Mandanas RY, et al. The effect of fluid loading, blood transfusion, and catecholamine infusion on oxygen delivery and consumption in patients with sepsis. *Am Rev Respir Dis*. 1986;134:873–878. [PubMed: 3777684]
128. Mink RB, Pollack MM. Effect of blood transfusion on oxygen consumption in pediatric septic shock. *Crit Care Med*. 1990;18:1087–1091. [PubMed: 2209035]
129. Sell LL, Cullen ML, Whittlesey GC, et al. Hemorrhagic complications during extracorporeal membrane oxygenation: prevention and treatment. *J Pediatr Surg*. 1986;21:1087–1091. [PubMed: 3491894]
130. Bjerke HS, Kelly RE Jr, Foglia RP, et al. Decreasing transfusion exposure risk during extracorporeal membrane oxygenation (ECMO). *Transfus Med*. 1992;2:43–49. [PubMed: 1308462]
131. McCoy-Pardington D, Judd WJ, Knafl P, et al. Blood use during extracorporeal membrane oxygenation. *Transfusion*. 1990;30:307–309. [PubMed: 2349629]
132. Plotz F. Extracorporeal membrane oxygenation and clotting revisited. *J Pediatr*. 1997;130:847–848. [PubMed: 9152304]
133. Green TP, Isham-Schopf B, Steinhorn RH, et al. Whole blood activated clotting time in infants during extracorporeal membrane oxygenation. *Crit Care Med*. 1990;18:494–498. [PubMed: 2328593]
134. Agarwal HS, Bennett JE, Churchwell KB, et al. Recombinant factor seven therapy for postoperative bleeding in neonatal and pediatric cardiac surgery. *Ann Thorac Surg*. 2007;84:161–168. [PubMed: 17588404]
135. Dager WE, Gosselin RC, Yoshikawa R, Owings JT. Lepirudin in heparin-induced thrombocytopenia and extracorporeal membranous oxygenation. *Ann Pharmacother*. 2004;38:598–601. [PubMed: 14982973]
136. Deitcher SR, Topoulos AP, Bartholomew JR, Kichuk-Chrisant MR. Lepirudin anticoagulation for heparin-induced thrombocytopenia. *J Pediatr*. 2002;140:264–266. [PubMed: 11865285]
137. Jamieson WR, Dryden PJ, O'Connor JP, et al. Beneficial effect of both tranexamic acid and aprotinin on blood loss reduction in reoperative valve replacement surgery. *Circulation*. 1997;96:II-96-II-100; discussion II100–II101.
138. Mejak B, Giacomuzzi C, Heller E, et al. Argatroban usage for anticoagulation for ECMO on a post-cardiac patient with heparin-induced thrombocytopenia. *J Extra Corpor Technol*. 2004;36:178–181. [PubMed: 15334762]
139. Verrijckt A, Proulx F, Morneau S, Vobecky S. Activated recombinant factor VII for refractory bleeding during extracorporeal membrane oxygenation. *J Thorac Cardiovasc Surg*. 2004;127:1812–1813. [PubMed: 15173742]
140. Wittenstein B, Na C, Pavo H, Goldman A. Recombinant factor VII for severe bleeding during extracorporeal membrane oxygenation following open heart surgery. *Pediatr Crit Care Med*. 2005;6:473–476. [PubMed: 15982438]

141. Shanley CJ, Hultquist KA, Rosenberg DM, et al. Prolonged extracorporeal circulation without heparin. Evaluation of the Medtronic Minimax oxygenator. *ASAIO J.* 1992;38:M311–M316.
142. Young G, Yonekawa KE, Nakagawa P, Nugent DJ. Argatroban as an alternative to heparin in extracorporeal membrane oxygenation circuits. *Perfusion.* 2004;19:283–288. [PubMed: 15506032]
143. Downard CD, Betit P, Chang RW, et al. Impact of AMICAR on hemorrhagic complications of ECMO: a ten-year review. *J Pediatr Surg.* 2003;38:1212–1216. [PubMed: 12891495]
144. Foland JA, Fortenberry JD, Warshaw BL, et al. Fluid overload before continuous hemofiltration and survival in critically ill children: a retrospective analysis. *Crit Care Med.* 2004;32:1771–1776. [PubMed: 15286557]
145. Hoover NG, Heard M, Reid C, et al. Enhanced fluid management with continuous venovenous hemofiltration in pediatric respiratory failure patients receiving extracorporeal membrane oxygenation support. *Intensive Care Med.* 2008;34:2241–2247. [PubMed: 18629472]
146. Lochan SR, Adeniyi-Jones S, Assadi FK, et al. Coadministration of theophylline enhances diuretic response to furosemide in infants during extracorporeal membrane oxygenation: a randomized controlled pilot study. *J Pediatr.* 1998;133:86–89. [PubMed: 9672516]
147. Santiago MJ, Sanchez A, Lopez-Herce J, et al. The use of continuous renal replacement therapy in series with extracorporeal membrane oxygenation. *Kidney Int.* 2009;76:1289–1292. [PubMed: 19794394]
148. Smith AH, Hardison DC, Worden CR, et al. Acute renal failure during extracorporeal support in the pediatric cardiac patient. *ASAIO J.* 2009;55:412–416. [PubMed: 19381083]
149. Brody JI, Pickering NJ, Fink GB, Behr ED. Altered lymphocyte subsets during cardiopulmonary bypass. *Am J Clin Pathol.* 1987;87:626–628. [PubMed: 3578136]
150. Schutze GE, Heulitt MJ. Infections during extracorporeal life support. *J Pediatr Surg.* 1995;30:809–812. [PubMed: 7666312]
151. Zach TL, Steinhorn RH, Georgieff MK, et al. Leukopenia associated with extracorporeal membrane oxygenation in newborn infants. *J Pediatr.* 1990;116:440–444. [PubMed: 2308038]
152. Piena M, Albers MJ, Van Haard PM, et al. Introduction of enteral feeding in neonates on extracorporeal membrane oxygenation after evaluation of intestinal permeability changes. *J Pediatr Surg.* 1998;33:30–34. [PubMed: 9473094]
153. Scott LK, Boudreaux K, Thalje F, et al. Early enteral feedings in adults receiving venovenous extracorporeal membrane oxygenation. *JPEN J Parenter Enteral Nutr.* 2004;28:295–300. [PubMed: 15449567]
154. Arnold JH, Truog RD, Orav EJ, et al. Tolerance and dependence in neonates sedated with fentanyl during extracorporeal membrane oxygenation. *Anesthesiology.* 1990;73:1136–1140. [PubMed: 2248393]
155. Yaucher NE, Fish JT, Smith HW, Wells JA. Propylene glycol-associated renal toxicity from lorazepam infusion. *Pharmacotherapy.* 2003;23:1094–1099. [PubMed: 14524641]
156. Frattalone J, Fuhrman BP, Thompson AE. Treatment of air leak during extracorporeal membrane oxygenation: total apneic lung rest. *Clin Res.* 1987;35:912A.
157. Keszler M, Subramanian KN, Smith YA, et al. Pulmonary management during extracorporeal membrane oxygenation. *Crit Care Med.* 1989;17:495–500. [PubMed: 2656095]
158. Kugelman A, Saiki K, Platzker AC, Garg M. Measurement of lung volumes and pulmonary mechanics during weaning of newborn infants with intractable respiratory failure from extracorporeal membrane oxygenation. *Pediatr Pulmonol.* 1995;20:145–151. [PubMed: 8545165]
159. Cronin J. *Cycling: An Alternative Method for Weaning for ECMO.* Breckinridge, CO: CNMC National ECMO Symposium; 1990:69.
160. Martin GR, Short BL. Doppler echocardiographic evaluation of cardiac performance in infants on prolonged extracorporeal membrane oxygenation. *Am J Cardiol.* 1988;62:929–934. [PubMed: 3177240]

161. Camboni D, Philipp A, Lubnow M, et al. Support time-dependent outcome analysis for veno-venous extracorporeal membrane oxygenation. *Eur J Cardiothorac Surg*. 2011.
162. Domico MB, Ridout DA, Bronicki R, et al. The impact of mechanical ventilation time before initiation of extracorporeal life support on survival in pediatric respiratory failure: a review of the extracorporeal life support registry. *Pediatr Crit Care Med*. 2011 Apr 7 (Epub ahead of print).
163. Adolph V, Ekelund C, Smith C, et al. Developmental outcome of neonates treated with extracorporeal membrane oxygenation. *J Pediatr Surg*. 1990;25:43–46. [PubMed: 1688940]
164. Matamoros A, Anderson JC, McConnell J, Bolam DL. Neurosonographic findings in infants treated by extracorporeal membrane oxygenation (ECMO). *J Child Neurol*. 1989(4 Suppl):S52–S61.
165. Hofkosh D, Thompson AE, Nozza RJ, et al. Ten years of extracorporeal membrane oxygenation: neurodevelopmental outcome. *Pediatrics*. 1991;87:549–555. [PubMed: 1707157]
166. Krummel TM, Greenfield LJ, Kirkpatrick BV, et al. The early evaluation of survivors after extracorporeal membrane oxygenation for neonatal pulmonary failure. *J Pediatr Surg*. 1984;19:585–590. [PubMed: 6502432]
167. Lott IT, McPherson D, Towne B, et al. Long-term neurophysiologic outcome after neonatal extracorporeal membrane oxygenation. *J Pediatr*. 1990;116:343–349. [PubMed: 2407817]
168. Towne BH, Lott IT, Hicks DA, Healey T. Long-term follow-up of infants and children treated with extracorporeal membrane oxygenation (ECMO): a preliminary report. *J Pediatr Surg*. 1985;20:410–414. [PubMed: 4045667]
169. Koumbourlis AC, Motoyama EK, Mutich RL, et al. Lung mechanics during and after extracorporeal membrane oxygenation for meconium aspiration syndrome. *Crit Care Med*. 1992;20:751–756. [PubMed: 1597027]
170. Boykin AR, Quivers ES, Wagenhoffer KL, et al. Cardiopulmonary outcome of neonatal extracorporeal membrane oxygenation at ages 10–15 years. *Crit Care Med*. 2003;31:2380–2384. [PubMed: 14501970]
171. Heulitt MJ, Moss MM, Walker WM. *Morbidity and Mortality in Pediatric Patients with Respiratory Failure*. Extracorporeal Life Support Meeting: Ann Arbor, University of Michigan; 1993:41.
172. Barrett CS, Bratton SL, Salvin JW, et al. Neurological injury after extracorporeal membrane oxygenation use to aid pediatric cardiopulmonary resuscitation. *Pediatr Crit Care Med*. 2009;10:445–451. [PubMed: 19451851]
173. Fajardo EM. Outcome and follow-up of children following extracorporeal life support. In: Zwischenberger JB, Bartlett RH, eds. ECMO: extracorporeal cardiopulmonary support in critical care. MI: University of Michigan; 1996:976–983.e3.
174. Lequier L, Joffe AR, Robertson CM, et al. Two-year survival, mental, and motor outcomes after cardiac extracorporeal life support at less than five years of age. *J Thorac Cardiovasc Surg*. 2008;136:976–983.e3. [PubMed: 18954639]
175. Fligor BJ, Neault MW, Mullen CH, et al. Factors associated with sensorineural hearing loss among survivors of extracorporeal membrane oxygenation therapy. *Pediatrics*. 2005;115:1519–1528. [PubMed: 15930212]
176. Ibrahim AE, Duncan BW, Blume ED, Jonas RA. Long-term follow-up of pediatric cardiac patients requiring mechanical circulatory support. *Ann Thorac Surg*. 2000;69:186–192. [PubMed: 10654511]
177. Linden VB, Lidegran MK, Frisen G, et al. ECMO in ARDS: a long-term follow-up study regarding pulmonary morphology and function and health-related quality of life. *Acta Anaesthesiol Scand*. 2009;53:489–495. [PubMed: 19226296]
178. Delmo Walter EM, Alexi-Meskishvili V, Huebler M, et al. Rescue extracorporeal membrane oxygenation in children with refractory cardiac arrest. *Interact Cardiovasc Thorac Surg*. 2011;12:929–934.
179. Grist G, Whittaker C, Merrigan K, et al. Defining the late implementation of extracorporeal membrane oxygenation (ECMO) by identifying increased mortality risk using specific physiologic cut-points in neonatal and pediatric respiratory patients. *J Extra Corpor Technol*. 2009;41:213–219. [PubMed: 20002075]

180. Herridge MS, Tansey CM, Matte A, et al. Functional disability 5 years after acute respiratory distress syndrome. *N Engl J Med*. 2011;364:1293–1304. [\[PubMed: 21470008\]](#)

181. Mehta NM, Turner D, Walsh B, et al. Factors associated with survival in pediatric extracorporeal membrane oxygenation—a single-center experience. *J Pediatr Surg*. 2010;45:1995–2003. [\[PubMed: 20920718\]](#)

182. Taylor AK, Cousins R, Butt WW. The long-term outcome of children managed with extracorporeal life support: an institutional experience. *Crit Care Resusc*. 2007;9:172–177. [\[PubMed: 17536987\]](#)

183. Thiagarajan RR, Brogan TV, Scheurer MA, et al. Extracorporeal membrane oxygenation to support cardiopulmonary resuscitation in adults. *Ann Thorac Surg*. 2009;87:778–785. [\[PubMed: 19231388\]](#)

184. Matsuwaka R, Matsuda H, Kaneko M, et al. Experimental evaluation of a heparin coated ECMO system simplified with a centrifugal pump. *ASAIO Trans*. 1990;36:M473–M475.

185. Palatianos GM, Dewanjee MK, Kapadvanjwala M, et al. Cardiopulmonary bypass with a surface-heparinized extracorporeal perfusion system. *ASAIO Trans*. 1990;36:M476–M479.

186. Roch A, Lepaul-Ercole R, Grisoli D, et al. Extracorporeal membrane oxygenation for severe influenza A (H1N1) acute respiratory distress syndrome: a prospective observational comparative study. *Intensive Care Med*. 2010;36:1899–1905. [\[PubMed: 20721530\]](#)

187. McMullan DM, Emmert JA, Permut LC, et al. Minimizing bleeding associated with mechanical circulatory support following pediatric heart surgery. *Eur J Cardiothorac Surg*. 2011;39:392–397. [\[PubMed: 20801051\]](#)

188. Custer JR. The evolution of patient selection criteria and indications for extracorporeal life support in pediatric cardiopulmonary failure: next time, let's not eat the bones. *Organogenesis*. 2011;7:13–22. [\[PubMed: 21317556\]](#)

McGraw Hill

Copyright © McGraw-Hill Global Education Holdings, LLC.

All rights reserved.

Your IP address is **132.174.254.28**

Access Provided by: Walter Reed National Military Medical Center

[Silverchair](#)